

INFORMING ENTREPRENEURS? INITIAL PUBLIC OFFERINGS AND NEW BUSINESS FORMATION*

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We examine the spillover effects of local initial public offerings (IPOs) on new business formation. Our findings reveal a substantial 1% to 4% increase in new business registrations following the introduction of new IPOs. These effects are particularly pronounced in counties facing higher economic uncertainty, highlighting the vital role of IPOs in conveying crucial information, whether through signaling potential success or providing direct information. Moreover, our research indicates that these effects extend beyond mere "signaling of success" and are largely driven by the valuable information transmitted by the IPO itself. Specifically, new business registrations are significantly influenced by the extent of Edgar downloads related to the IPO firm's public disclosures and the informational content within the IPO firm's S-1 disclosure. A comprehensive field survey further supports these conclusions, affirming the presence of information spillovers from local IPOs that positively impact new business formation.

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I. INTRODUCTION

The study of initial public offerings (IPOs) and their effects over the last few decades has been substantial. This extensive body of literature initially focused on the effects of IPOs on the IPO company's business, and has documented various consequences of going public, including a decline in the IPO firm's innovative output, (Bernstein, 2015), changes in mergers and acquisitions (Celikyurt et al., 2010), wages (Babina et al., 2020) and employment growth (Borisov et al., 2022). More recently, the focus of research has shifted towards exploring the potential spillover effects of IPO activities on the broader economy, encompassing aspects such as labor market movement, local economic effects, and industry dynamics (Butler et al., 2019; Aghamolla and Thakor, 2022; Cornaggia et al., 2022). In this paper, we delve into a novel dimension of potential IPO spillovers, namely on local entrepreneurship.

Studying the spillover effect of IPOs on entrepreneurial activity is of first-order interest. New business formation plays a pivotal role in economic development (Haltiwanger et al., 2013). Extensive research using U.S. Census data consistently underscores that new firms drive job growth (Davis and Haltiwanger 1992; Davis et al. 1998; Haltiwanger, Jarmin, and Miranda 2013; Decker et al. 2014; Fairlie, Miranda, and Zolas 2019). At the same time, however, recent studies indicate a secular decline in the rate of business dynamism and new firm creation (Decker et al., 2017; Hathaway & Litan, 2014). Understanding what drives entrepreneurial entry and in turn the process of new business formation is thus of paramount importance.

Importantly, newly listed public firms generate and disclose substantial amounts of information. This information serves as both a signal of success prospects for companies in the region or industry, as well as a source of direct incremental information provision through firm disclosures, which may include analysis of the market and competitive dynamics within the

industry, financial performance, expected future outlook, current, and future investment outlays, material contracts, and business risks, all of which may be informative to would-be entrepreneurs considering the launch of a new company. Additionally, the discussion and analysis of such disclosures by information intermediaries like financial analysts and the business press further generate information for aspiring entrepreneurs. Collectively, the information generated by an IPO can enhance the information environment within the industry vertical and geographical area of publicly-traded firms, reducing uncertainty about demand, supply, and cost conditions (e.g. Lowry and Schwert, 2002; Badertscher et al., 2013), thus increasing the probability of new business success.

Importantly to our study, entrepreneurs are actively involved in the process of identifying profit opportunities to launch successful and sustainable businesses. These opportunities may be related to the production's output or input side, or a combination of both. Information seeking and interpretation are at the core of this entrepreneurial process, making information gathering a crucial activity for prospective entrepreneurs when endeavoring to start a new business venture. We hypothesize that the information generated by an IPO—whether signaling success prospects or as a result of direct firm disclosures and their dissemination—serves to reduce uncertainty regarding the future returns of entrepreneurial activities, ultimately fostering entrepreneurial entry following IPO events.

To empirically test this hypothesis, we leverage geographic variations in IPOs across U.S. counties and time. Our analysis incorporates data on new for-profit business registrations in the United States, offering zip code and county-level information from the Startup Cartography Project (SCP). Recognizing that soft information complements and adds color to the publicly transmitted information in disclosures, with this transmission likely diminishing with distance

(Liberti and Petersen, 2019), we anticipate that the effect of an IPO on new business formation will be more pronounced the closer the geographic proximity to the IPO firm. We therefore employ a difference-in-differences (D.D.) design comparing new business launches in counties with recent IPOs to those in counties that have not experienced a recent IPO, both before and after the occurrence of the IPO. Our D.D. specification includes a vector of time-varying county characteristics (population, income per capita, and the number of public firms) as well as county and quarter-year fixed effects. Using our D.D. specification, we document a 1-4% increase in new business registrations in the quarter following the IPO of a firm in the county. This translates into an average of 4-15 new business registrations per quarter after the IPO.

An obvious consideration in interpreting these estimates is the possible effect of omitted factors, such as local economic growth in the area, which may jointly influence IPO activity and new business formation. Prior research offers mixed results regarding the impacts of IPOs on local economic conditions. While some studies suggest positive effects of IPOs on home price growth, retail establishments, and gentrification (Butler et al., 2019), more recent work finds that IPOs do not necessarily lead to growth in employment, establishment growth, and population (Cornaggia et al., 2022). To mitigate these concerns, we directly examine the relation between IPOs and economic growth. We find that neither contemporaneous growth in wages nor employment appear to rise in the county following an IPO in our sample. Furthermore, we find evidence of parallel trends in our D.D. estimation, further alleviating concerns that differential patterns in growth in entrepreneurial entry are driving our results.

Given the mixed evidence and the endogenous nature of IPOs, we then employ an identification approach in the vein of Bernstein (2015) to causally assess the effects of IPOs on entrepreneurship. The central challenge in this endeavor lies in the need to identify the impact

of going public on entrepreneurship, while recognizing that prosperous local economies are more inclined to host the types of firms that eventually go public and exhibit robust new business formation. To address this inherent selection issue, we leverage quasi-random variation in whether a firm completes its IPO, thereby allowing us to isolate the causal effect of going public.

To isolate this quasi-random variation, we follow a methodology motivated by Bernstein (2015), and use the returns on the NASDAQ index in the months immediately following a firm's IPO filing as an instrumental variable for IPO completion. The rationale behind this instrument is grounded in the dynamics of the initial months after filing for an IPO, known as the book building phase. During this period, a firm must actively generate investor interest in its stock. Market fluctuations during the book building phase can significantly influence investors' appetite for the firm's stock and, in turn, affect the likelihood of the IPO being completed. Importantly, macro-level stock market fluctuations during this phase are unlikely to have a lasting impact on the long-term economic activities of the county where the IPO occurs. In other words, the NASDAQ index serves as an aggregate shock that remains consistent at the county-quarter level, making it less susceptible to the influence of local conditions in a specific county. This isolation allows us to examine the effects of local IPOs more precisely. Using the instrumented IPOs, we continue to observe a similar positive relationship between IPOs and new business registrations. For additional robustness, we conduct an entropy balancing matching analysis to account for the potential differences between the treatment and control counties, with similar findings.

We next assess whether individuals are actively exploring the potential of initiating new businesses following a local IPO. We examine whether an IPO filing may prompt individuals

to consider the prospects of launching a business, even if they ultimately decide against it. To gauge this, we measure “entrepreneurial interest” using Google search shares for terms related to entrepreneurship, such as “how to start a business” or “how to incorporate” (Barrios et al., 2020). Consistent with our main findings, the D.D. specification documents a significant and substantial increase (12%) in the likelihood of a county ranking in the top quartile of search share activity related to entrepreneurship and new business formation terms following an IPO. Moreover, as we would expect, the impact of local IPOs on new business formation is considerably more pronounced in counties characterized by high level of entrepreneurship-related internet search activity. The findings confirm that the observed uptick in business registrations is correlated with increased interest in entrepreneurship, as evidenced by the increased online searches for entrepreneurship-related terms.

Next, we delve into the mechanisms underlying our primary findings. Our hypothesis centers on the idea that the relation between IPOs and new business formation is, in part, driven by the IPO’s informational role in reducing uncertainty. This can occur through two potential channels: one in which the IPO signals the prospects of success for new businesses (“signal of success”), and the other involving the actual information content generated by the disclosures of the new firm and the associated transfer of soft information (“direct information”).

We begin our subsequent tests by examining this general relationship. If information from the IPO reduces uncertainty about new business prospects, we would expect our results to be concentrated in situations where economic uncertainty is the highest ex-ante. Consistent with this reasoning, we observe that the positive link between IPO and local entrepreneurship is more pronounced in locations where ex-ante economic uncertainty, measured by wage growth volatility, is notably high.

We then further investigate cases when an IPO is likely to provide more versus less information. The production of new information is more likely when a private firm goes public, as opposed to a subsidiary being spun off from a public company, since the parent firm already discloses public information about the subsidiary before the IPO. Consistent with our hypothesized information mechanism, we observe that the increase in new firm registrations is primarily concentrated around private-to-public IPOs. While these empirical patterns strongly suggest the presence of an information channel of some form, they do not in and of themselves allow us to differentiate conclusively between the “signal of success” and the “direct information” channels.

To delve deeper into the importance of the direct information channel, we begin by examining how the volume of Edgar downloads of public firm reports changes around new IPOs, and how these downloads relate to new business registrations. One implication of the direct information hypothesis is that we should expect heightened information acquisition patterns following an IPO event. Our analysis reveals that information consumption, as measured by downloads of public filings from Edgar, is notably higher in counties that experience an IPO. Further, we find that counties actively engaging in downloading these filings also exhibit higher rates of new business registrations. Notably, the positive relationship between an IPO and new business formation is magnified in counties with high levels of S-1 downloads. Additionally, it is more pronounced when the S-1 forms filed by the IPO firms are richer in information content, as determined by factors such as length, complexity, and the extent of content related to entrepreneurial activities. These findings collectively support the hypothesis that the positive link between IPOs and new business registrations is at least partially driven by the acquisition of direct information from the IPO firm's public disclosures.

Finally, to complement the observational data on information spillovers from IPOs, we conducted a field survey of entrepreneurs, involving 503 participants, consisting of both seasoned entrepreneurs and those aspiring to enter the entrepreneurial realm. A majority of these respondents held a college degree or higher qualification, representing a diverse array of industries. Our survey analysis enables us to gain insights from firsthand responses into the mechanisms through which local IPOs affect and inform entrepreneurial activity. This builds on emerging accounting research that leverages surveys to glean deeper insights about studying underlying mechanisms that explain patterns detected in archival data (Bernard et al. 2018; Hanlon and Shroff, 2022; Call et al. 2023; Leonelli et al. 2023).

The survey yielded noteworthy findings, with respondents reporting awareness of an average of 2.07 IPOs within their respective industries over the past five years. Importantly, the survey responses revealed a significant trend—many respondents either maintained a neutral stance or expressed an increased inclination to initiate a new entrepreneurial venture when informed about a local IPO. Nearly half of the respondents conveyed that the information acquired from local IPOs had a positive influence on their decision to pursue entrepreneurship, emphasizing the value of this information acquisition compared to the IPO events themselves.

The survey also shed light on the sources of information that entrepreneurs relied on when seeking insights from local IPOs. Their primary sources included traditional media, the official websites of the IPO companies, SEC filings, and various social media platforms. These findings serve to complement our previous analyses and collectively support the notion that local IPOs play a pivotal role in providing valuable information that catalyzes entrepreneurial endeavors, with entrepreneurs actively seeking guidance on various aspects such as product market conditions, operational strategies, technologies, valuation, and potential funding opportunities.

In sum, our empirical analyses combined with our field survey results collectively confirm the presence of an information-based impact of IPOs on entrepreneurial activity. They strongly underscore that this effect is driven by the acquisition of direct information stemming from the IPO process, and not merely by a “signal of success.” We note that while our findings emphasize the significance of this information disclosure mechanism, they do not exclude the possibility of other contributing factors. Nonetheless, our results provide robust and compelling support for the pivotal role of information disclosure in the connection between IPOs and the stimulation of new business formation.

This study makes several contributions to the literature. First, our work contributes to the extensive existing literature on IPOs, and in particular, IPO spillovers to the real economy (see, e.g., Cornaggia et al., 2018; Butler et al., 2019; Aghamola and Thakor, 2022). We provide new, causal evidence on the effects of IPOs on local entrepreneurial activity, and the importance of information spillovers in driving these effects.

Second, our findings also provide a first insight into the process through which potential entrepreneurs obtain relevant information when deciding to launch a new venture. Our paper contributes to existing research on the externalities from public firm disclosures (Durnev and Mangen, 2009; Beatty et al., 2013; Badertscher et al., 2013; Shroff et al., 2014; Bernard et al., 2020). For example, Badertscher, Shroff, and White (2013) show that private firms are more responsive to their investment opportunities when they operate in industries with greater public firm presence. Durnev and Mangen (2009) show that accounting restatements are associated with lower abnormal returns and reduced investment by non-restating firms in the industry. The authors suggest a “learning” effect in that restatements convey information about investment prospects to the managers of restating firms’ competitors. While these studies primarily focus

on the intensive margin effects of disclosure spillovers—the investment decisions of established firms, our study contributes to this literature by using IPOs and the broader entrepreneurial setting of new business formation to show that public firm information disclosures have positive externalities that also affect the extensive margin of new business formation. Our findings provide a more holistic understanding of how information spillovers from public firms may spur entrepreneurial activity and new firm formation.

Closest to our work may be Breuer (2021), who reveals the facilitative role of European mandatory financial reporting regulations in promoting market entry and product market competition. Our findings complement Breuer (2021), while offering a unique contribution. While both studies delve into industrial dynamics and the influence of disclosure, our work focuses on the micro determinants of entrepreneurial entry. This distinction places significant emphasis on our specific focus on new business formation, which, in turn, enables a more detailed exploration of entrepreneurial dynamics, particularly within the realm of innovation. Furthermore, our study highlights the active involvement of entrepreneurs and aspiring entrepreneurs in the acquisition of information from IPO filings and Google searches, accentuating the significance of information channels in our context. An important distinction lies in the source of information spillover, which in our study predominantly originates from IPO filings in an environment where information disclosure for private firms is not required and therefore would-be entrepreneurs face a higher level of information frictions. In contrast, Breuer (2021) focuses on mandated financial statement disclosure from private firms in an environment where information is more widely available. This distinction in the source of information constitutes a pivotal aspect of our research, as we examine spillovers extending beyond the industry and into areas beyond direct competitive environments. Additionally, our

US-based sample provides direct evidence of entrepreneurial activity within the United States, potentially offering distinct insights when compared to the European context explored in Breuer's work. We view the results from these two papers as complementary and informative of the differences between the two settings.

Building upon the prior line of reasoning, our paper also makes a significant contribution to the ongoing discussion surrounding the costs and benefits of disclosure regulation. Despite its widespread application, the justification for disclosure regulation can be a challenging task due to the inherent market-based incentives for firms to disclose information (Admati and Pfleiderer, 2000; Leuz and Wysocki, 2016; Berger, 2011). This stems from the firm's natural inclination to avoid the costs associated with withholding information, thereby motivating them to disclose. One rationale supporting mandatory disclosure is the existence of positive externalities stemming from such disclosure (e.g., Breuer et al., 2020; Kim and Valentine, 2021; Kim and Valentine, 2022). In this context, our paper offers additional evidence that aligns with the notion of positive externalities arising from corporate disclosures, particularly in their role in stimulating new business formation.

II. DATA SOURCES AND KEY VARIABLES

In this section, we describe our variable and the various databases used to construct them. Overall, our sample comprises 359,892 county-quarter observations. When data is available, we also conduct tests at the county-industry-quarter level, with a sample comprising 866,359 county-industry-quarters in additional tests further described below.

A. Initial Public Offerings

Our main proxy for the availability of public disclosures relevant for entrepreneurs is *Post IPO*, defined as a county-specific indicator variable that equals to one for every county-quarters after the IPO quarter within a 3-year treatment window, and zero otherwise. IPOs represent a significant expansion in public information available to the public through mandated corporate filings, newspaper coverages, and analyst reports, all of which provide relevant information for would-be entrepreneurs. For example, Ryans (2017) shows that the most downloaded corporate filings are the S-1 filings from IPOs—even more so than 10-Ks, 10-Qs, and 8-Ks. This is a testament to the importance of IPO-related disclosures as a source of information to the public, and by extension to potential entrepreneurs.

We include counties that never had an IPO in our sample. These serve as a benchmark comparison group and have the variable *Post IPO* set to zero over the entire sample period. We collect IPO issuance data using Refinitiv's SDC New Issues database. The sample starts from 1988 Q1 and ends in 2016 Q4. The data contains county as well as industry information at the 2-digit SIC level for each IPO, which we exploit in certain model specifications described further below. Following the IPO literature, we exclude IPO filings of financial firms (SIC codes between 6000 and 6999), unit offerings, closed-end funds (including REITs), ADRs, limited partnerships, special acquisition vehicles, and spinoffs. We identify 7,829 IPOs over the sample period of 1988 to 2016. In Figure 1, we plot the geographic variation in IPOs among U.S. counties. We observe variation across geography and time with regards to IPO activity.

B. Business Formation Measures

We use two complementary measures of entrepreneurial activity, which together serve as our dependent variable of interest.

II.B.1. Business Registrations

Our first measure of new business formation uses new business registrations. Our business registration data is obtained at the county level from the Startup Cartography Project (SCP) database. The SCP leverages business registration records, which are public records created when an individual registers a new business as a corporation, LLC, or partnership. Importantly, as noted by Guzman and Stern (2020), while it is possible to found a new business without business registration (e.g., a sole proprietorship), the benefits of registration are substantial, and include limited liability, various tax benefits, the ability to issue and trade ownership shares, and credibility with potential customers. Business registrations reflect the population of incorporated businesses operating in a location (which may differ from their state of incorporation) that have taken on a form that is a pre-requisite for growth or employment. We define our main variable of interest, $\text{Log}(1 + \text{New Business Reg})$, as the natural logarithm of one plus the number of new business registrations in the county in the period of interest.

The SCP business registration data covers 50 states and territories (e.g. the District of Columbia) from 1988 to 2014 and 47 after 2014 (IL, SC, and MI drop out of the sample after 2014). Data is aggregated to the county level. A limitation of the data is that it does not allow for identification of the firms' industry. All in all, $\text{Log}(1 + \text{New Business Reg})$ complements $\text{Log}(1 + \text{Establishment Birth})$ (introduced in the next section) by providing a more refined measure of new business formation. Specifically, $\text{Log}(1 + \text{New Business Reg})$ avoids the limitation of the establishment birth measure, which may be driven by new branches of existing companies rather than new business formation. We graph the county variation of *New Business Reg* in Figure 2.

II.B.2. Establishment Births

Our second measure comes from the U.S. Census Statistics of U.S. Business database (SUSB), which covers all U.S. business establishments with paid employees spanning the years 2000 to 2015. The data source provides information on new establishment births at the county level at the 2-digit NAICS level. The SUSB defines an establishment birth as establishments that have zero employment in the first quarter of the initial year and positive employment in the first quarter of the subsequent year. We define $\text{Log}(1 + \text{Establishment Birth})$ as the natural logarithm of one plus the number of new establishment formations, measured at the county and 2-digit NAICS level.

C. Public Disclosure Download Data

To directly examine the consumption of public firm disclosures, we retrieve the server logs associated with the SEC's Edgar website. We define $\text{Log}(1 + \text{Downloads})$ as the natural logarithm of one plus the total number of downloads in a given county-quarter. The SEC's server log provides the Internet Protocol (I.P.) address (anonymized), a timestamp, and the physical location of the I.P. address. Our dataset covers the period from 2005 to 2017.

To examine the channel and outcome of consuming public firm information at the entrepreneurial firm level, we obtain a large sample of start-ups from Crunchbase, a crowd-sourced database that tracks start-ups. Crunchbase describes itself as "the leading platform to discover innovative companies and the people behind them." The primary source of data in Crunchbase is TechCrunch, an online publisher of technology industry news. Both TechCrunch and Crunchbase were founded in 2005 and include backfill data from the mid-1990s. Our sample covers the period from 2005 to 2019. We require start-up companies to be in the U.S.

and have non-missing values in their geographic location, industry classification, age, and the number of employees. We have 227,310 unique start-up companies.

We perform a manual matching process between Crunchbase start-ups and Edgar Server Logs using IP addresses. This process begins by obtaining a comprehensive file listing from the SEC and extracting all relevant file URLs. We subsequently use the URLs to download the Edgar IP log files. Within the Edgar Server Logs, we have access to IP addresses and information regarding the public firms associated with each IP address search. To enhance the accuracy of this information, we cross-reference these IP addresses with the MaxMind Data service. MaxMind Data provides us with location and organizational (company) details for each IP address.

Next, we match the company names obtained from MaxMind with the company names listed in Crunchbase start-ups. Among the 12,717 firms within the Edgar downloads that have valid company names, we successfully match 1,444 firms. Subsequently, we utilize the location information sourced from Crunchbase to calculate both the number of Edgar downloads originating from these startups and the count of startups implementing Edgar downloads at the county-year level. Additional summary statistics for this subsample can be found in Online Appendix Table OA5.

D. Summary Statistics

Table 1 Panel A provides descriptive statistics for our key variables at the county-quarter level. The average county-quarter in our sample has 371 new business registrations. The average annual income per capita in the sample is \$26,670. Columns (2) and (3) show that counties with at least one IPO are observably different from counties with no IPOs. These

differences motivate us to include county fixed effects as outlined in the next section. We also include income per capita, population, and the proportion of public firms as controls to account for these heterogeneities.

III. MAIN ANALYSIS

In this section we describe our empirical design as well as the results of our estimation of the relation between IPOs and local new business formation and entrepreneurial interest. We then conduct tests to rule out the alternative story that economic growth is driving our results, rather than information from public disclosure.

A. *IPOs and New Business Formation*

We begin by exploring the base relationship between IPO activity in a county (e.g., a county-headquartered firm going public) and new business formation as measured by the SCP database of new business registrations. We estimate a generalized difference-in-differences (D.D.) model. Because the SCP data does not provide information about the business' industries, our model's variation is across counties. Specifically, we estimate the following generalized difference-in-differences model with staggered treatment:

$$\text{Log}(1 + \text{New Business Reg})_{c,q} = \delta \text{Post IPO}_{c,q} + \alpha_c + \gamma_q + \beta' X_{c,q} + \theta_c q + \varepsilon_{c,q} \quad (1)$$

where we denote county by c and quarter by q . $\text{New Business Reg}_{c,q}$ is the number of new business registrations as reported in the SCP database. Standard staggered D.D. design would treat $\text{Post IPO}_{c,q}$ an indicator variable that equals one for the county-quarters after an IPO and zero otherwise. Such design cannot account for two important features of IPOs that might affect

our inference. First, many IPOs occur in just a few counties, leading to limited variation in the $Post\ IPO_{c,q}$ variable. Consequently, the interpretation of the coefficient on $Post\ IPO_{c,q}$ becomes challenging, as it only captures the effect of the first IPO in each county. Furthermore, for those counties that had IPOs before our sample period, $Post\ IPO_{c,q}$ is set to one only after an IPO within our sample period, making our results potentially sensitive to the choice of the sample period.

To address these concerns, we treat each IPO as a separate event and use a shorter event window. We construct our sample by gathering all IPO events together, and the $Post\ IPO_{c,q}$ variable is an indicator variable that equals one during the 3-year period after an IPO, and zero during the 3-year period before an IPO. We use the never-treated county quarters as a control group. The unit of observation remains at the county-quarter level, and we use weighted least square (WLS) to account for county-quarters that have multiple IPOs (so that we do overweight such county-quarters in our regressions). This research design overcomes the aforementioned concerns by ensuring that every IPO contributes to identifying variation in our treatment variable ($Post\ IPO_{c,q}$), and it eliminates sensitivity to the starting point of our sample period since we focus on a short window around each IPO event.¹

We include county fixed effects α_c and quarter-year fixed effects γ_q . We also include several control variables to capture time-varying economic conditions in the county such as the

¹ In Online Appendix Table 6, we show that our results are robust to a complete stacked design following Cengiz et al. (2019), where we limit the treatment sample to only those with either no or only one IPO within 6-year period of a county-quarter (i.e., no overlapping treatment window with the other IPOs) (Baker et al., 2022; Barrios, 2021). This much more restrictive sample reduces IPO events from 7829 to less than 500. Our results are also robust to an alternative “complete stacked design” specification where we match each IPO treatment event with all never-treated control counties during the treatment window (results untabulated).

log of county population and county income per capita. We cluster the standard errors at the county level. This design compares changes in new business registrations in county quarters where an IPO has been initiated to business registrations in county quarters without an IPO.

Table 2 reports the results of estimating various permutations of equation (1). In columns (1) and (2) our main dependent variable is $\text{Log}(1 + \text{New Business Reg}_{c,q})$ while in column (3) we use $\text{Log}(\text{New Business Reg}_{c,q})$ as our primary dependent variable and restrict the sample to non-zero observations. We find that areas that experience an IPO see an increase of 1-4% in new business registrations in the 12 quarters after the IPO (even after controlling for time-varying country economic conditions with county-year fixed effects). This increase translates to around 4 to 15 new business registrations on average in the IPO county in the post IPO quarters relative to non-IPO quarters.

Barrios (2021) states that one of the fundamental assumptions of the staggered-entry D.D. model is the parallel trend assumption, which states that in absence of the treatment (IPO in the county), the dependent variable of both the treatment and the control groups should exhibit the same trend. We thus assess graphically whether the parallel trends assumption holds in Figure 3. Specifically, Figure 3 graphically presents the D.D. estimates (with each dot representing annual-coefficients) in event time. In both panels, the counterfactual treatment effects in the pre-IPO periods are statistically indistinguishable from zero, providing further support for our inferences (parallel trends in the pre-period). Post-IPO, in contrast, we see a clear increasing treatment effect.

Of course, it is possible that spillover effects between geographically proximate counties may contaminate a subset of our control group. Since our treatment is at the geographic level, the question is what constitutes an independent geographic region. In our setting, the concern

is that a neighboring county may not be independent of activity in its neighboring counties with respect to information transmission. For example, to the extent that the informational benefits of a given county's IPOs (the treatment group) positively affect the entrepreneurial activity of a neighboring county, which in our specification is included in the control group, our estimates may be biased downward. We explore this issue in Table OA1 in the Online Appendix. We find that our results are stronger when we exclude the treated county's immediately neighboring counties, suggesting that our findings in Table 2 are likely conservative. Neighboring counties are likely experiencing some amount of the IPO's spillover effects. Because they are included in our control group, their inclusion will attenuate our estimates. Additionally, we find that inter-county spillover effects dissipate rapidly with distance. We find muted effects when examining counties that are "two-degrees removed" from the treatment county, suggesting that our control group in general (except for the closest neighbors) likely satisfies the SUTVA assumption.

While the difference-in-differences setup we employ uses staggered arrival of IPOs in the county to identify the effects on new business formation, a reasonable concern is that the emergence of IPOs in a county may not be exogenous but may relate in some way to county-specific attributes that vary over time and which also influence the formation of new businesses. While the IPO event typically signals that a firm has reached a stage in which its business is no longer likely to be limited to its local headquarters market, we nevertheless provide analysis that addresses the endogenous nature of IPOs (Aghamolla and Guttman, 2021).²

² We bolster our main analysis by employing entropy balancing. This approach mitigates any underlying disparities between the treatment and control counties. We report the outcomes of this analysis in Online Appendix Table 2, with the matching process relying on three control variables: Log Population, Log Income Per Capita, and Log

To allow for a clearer inference of the causal impact of an IPO completion on new business formation, we adopt an approach inspired by Bernstein (2015). Specifically, we leverage the occurrence of IPOs that are initially filed but later withdrawn due to sudden shifts in market conditions. In line with Bernstein’s methodology, we employ an instrumental variable for IPO completions at the firm level. Our instrument is based on the NASDAQ return in the two months following the IPO filing date, which allows us to calculate firm-specific probabilities of completing an IPO based on the NASDAQ returns following the filing. To estimate these probabilities, we perform the following first-stage regression:

$$IPO_{i,c,q} = \alpha + \beta NASDAQ\ Returns_{i,c,q} + \gamma X + \alpha_c + \gamma_q + \varepsilon_{c,q} \quad (2)$$

for filer i in country c and quarter q . IPO is the dummy variable of interest, indicating whether a filer goes public or remains private. $NASDAQ\ Returns$ are two-month returns after the IPO filing date. We include county fixed effects α_c and quarter-year fixed effects γ_q . We include a vector of control variables, X , which include the log of county population and county income per capita. We cluster the standard errors at the county level.

Our instrumental variable, the NASDAQ return, has two crucial features that contribute to our identification strategy. First, as a macro-level variable, it captures market conditions that can influence the decision to launch an IPO, but, importantly, it does not vary at the county quarter level. The IV is thus minimally impacted by local conditions, allowing us to isolate the

Num Pub Firm. Notably, the results from this entropy balancing matching analysis confirms the consistency of our core findings, adding further support to our conclusions.

effects of local IPOs. Second, by measuring the NASDAQ return over a two-month period following the IPO filing, we account for within-quarter variation across IPO filings in different counties. In this way, variation of the IV is meant to capture not only the completion of the IPO but also the availability of newly-disclosed information by the IPO firm, such as S-1 filings, 10K reports, 10Q reports, as well as other relevant market information like media articles and analyst reports.

Panel A of Table 3 presents the estimates from the first stage of the IV. As in Bernstein (2015), NASDAQ returns are significant and strong predictors of an IPO completion. In our sample of IPO filing firms, 78.6% of firms complete their IPO. Moreover, the F-statistic of 41.95 exceeds the conventional threshold of $F = 10$ and suggests that the instrument is strong and unlikely to be biased toward the OLS estimates (Bound, Jaeger, and Baker, 1995; Staiger and Stock, 1997). The second-stage equation estimates the impact of IPO on new business registration:

$$\text{Log}(1 + \text{New Business Reg})_{c,q} = \delta \widehat{IPO}_{i,c,q} + \alpha_c + \gamma_q + \beta' X_{c,q} + \varepsilon_{c,q} \quad (3)$$

where $\widehat{IPO}_{i,c,q}$ is the predicted values from the first-stage regression.

The estimates are reported in Panel B of Table 3. Consistent with the difference-in-differences estimation in the preceding table, we observe a positive and statistically significant coefficient on the instrumented IPO variable with similar magnitude as in the non-instrumented models. The estimates suggest a strong causal relationship between a new IPO in the county and subsequent new business formation.

To strengthen our causal interpretation, we augment the Bernstein design with an alternative shift-share-based IV strategy, building upon the methodology overviewed by Breuer (2022). This strategy leverages nationwide IPO waves across various industries and assesses their respective impacts on each county's exposure to these industries. The underlying concept is that when a surge in nationwide IPOs occurs within a specific industry, counties with a higher level of exposure to that industry are more likely to observe an IPO in that particular year.

This shift-share design harnesses both cross-sectional and time-series variation. In the cross-section, we account for each county's initial exposure to different industries in the pre-sample period. County i 's exposure to industry j is calculated as the total number of establishments in industry j within county i , divided by the total number of establishments across all industries in county i . This measure of industry exposure is derived from 1987 data, specifically one year before the commencement of our primary sample. Simultaneously, the time-series variation stems from an industry's fluctuations in the number of nationwide IPOs over time. Our IV is constructed by merging these two sources of variation and aggregating the data for each county-year across industries. We utilize this IV to instrument for the number of IPOs in a county-quarter within a 2SLS regression. The results, untabulated, reveal a consistently positive and significant association between the number of IPOs and new business formation, thus further reinforcing our key findings.

B. IPOs and Establishment Births

Next, we shift our focus to establishment birth as an alternative proxy for entrepreneurial activity, drawing upon the SUSB data outlined in Section III. A key advantage of using establishment births is that it allows us to observe the establishment's industry information. To

investigate whether IPO activity impacts establishment births, we conduct the following regression:

$$\text{Log}(1 + \text{Establishment Birth})_{c,i,t} = \delta \text{Post IPO}_{c,i,t} + \alpha_c + \mu_i + \gamma_t + \beta' X_{c,t} + \varepsilon_{c,i,t} \quad (4)$$

with observations measured at the county c , industry i , and year t level. $\text{Establishment Birth}_{c,i,t}$ is the number of new establishment births in the county-industry-year, as taken from the Census database. We include county α_c , industry μ_i , and year γ_t fixed effects. Finally, $X_{t,c}$ is a vector of time-varying county-specific control variables, including the log of population and income per capita. We cluster standard errors at the county level. Our coefficient of interest, δ , captures the percentage increase in entrepreneurial activity subsequent to an IPO in the same county-quarter and industry.

Table 4 Panel A provides the results of estimating equation (4). Column (1) demonstrates a positive link between $\text{Log}(1 + \text{Establishment Birth})_{c,i,t}$ and $\text{Post IPO}_{c,i,t}$. We find an 4% increase in the number of establishment births for an industry-year-county in which an IPO happens. Columns (2) similarly shows that this relation is robust to using $\text{Log}(\text{Establishment Birth})_{c,i,t}$ as the dependent variable, where we restrict the sample solely to observations with positive establishment births. In Panel B, we add another variable in the same regression specification that captures IPOs not in the focal industry. Since information spillover should be concentrated among firms in the same industry, we anticipate that the effects would be significantly lower for new businesses operating different industries of the IPO firm. Consistent with this expectation, we find no result when examining the effect of IPOs on business formation in other industries.

To examine the cumulative effect of IPOs, we consider public firm presence, defined as the proportion of public firms in each NAICS 3-digit industry-year, following Badertscher et al. (2013) and Shroff et al. (2017). The measure can be viewed as a cumulative measure of IPOs. Data on the total number of firms in each industry is obtained from the Census Bureau, and we proxy for the number of public firms in each industry using data from Compustat. To minimize the influence of outliers, we sort Public Firm Presence into quintiles each year to create the variable Pub Firm Presence Quintile. We present this analysis in Table OA3 in the Online Appendix. Consistent with our IPO findings, we find qualitatively similar results with this measure of public firm presence, furthering the notion of an information channel mechanism.

C. Alternative Measures of Entrepreneurial Activity: Google Searches

Next, we turn to an alternative measure of entrepreneurship, entrepreneurial interest, which captures the general interest in entrepreneurial activity, even if not acted upon. Specifically, we focus on internet searches for terms and phrases that directly relate to launching a business—what is referred to in the literature as entrepreneurial interest (Barrios et al., 2020).³ This measure provides us with a timelier response to spillovers from the IPO by examining whether IPO activity in the county leads individuals to begin contemplating entrepreneurship.

Table 5 employs a linear probability model to estimate a variation on equation (1) at the DMA level. The outcome is an indicator variable that is defined as one in the case that the DMA

³ We track trends for searches for these terms using the Google Health Trends API for all Nielsen Designated Market Areas (DMAs) at monthly frequency from January 2004 to December 2016. We aggregate the data to the quarter level and match the DMAs to Census incorporated places using a crosswalk provided by Nielsen. Specifically, we use the terms: “start a business,” “start your own business,” “start a company,” “how to incorporate,” “entrepreneurship,” “become an entrepreneur,” and “small business loan.” See Barrios et al. (2020) for more details.

is in the top quartile of entrepreneurial search in that quarter. Each specification includes controls for population, per capita income, and the number of public firms headquartered in the area. Our variable of interest is the *Post IPO* variable, which captures exposure to public firm information being produced by the IPO. The specification includes year and state fixed effects. Column (1) of Panel A indicates that after an IPO, the affected DMA is more likely to be in the highest quartile of entrepreneurial search activity. In Column (2), we further confirm that the result continues to hold when we instrument *Post IPO* with the NASDAQ return IV.⁴ In terms of economic magnitude, the 3% increase in the probability translates to a 12% increase over the base probability of 25%. Finally, in Panel B, we find that the impact of local IPOs on new business formation is significantly more pronounced in counties with high entrepreneurship-related searching activities.⁵

D. Are We Merely Picking Up Economic Growth?

An obvious concern is that our results above may be driven by concurrent economic growth in the county, that in turn drives new business formation. The existing literature has mixed results regarding the potential presence of positive spillovers from IPOs into the real economy—often referred to as an IPO “multiplier effect.” On one hand, IPOs have the potential to contribute to aggregate local economic growth by generating a wealth and liquidity shock

⁴ In untabulated analysis, we confirm these results continue to hold if we measure entrepreneurial search share as a continuous variable.

⁵ To provide additional evidence of spillover effects on entrepreneurial activity, we investigate two additional indicators related to new business activity and their financing. Specifically, we analyze loan data from the Small Business Administration (SBA) and venture capital data from Crunchbase. Our findings align with our expectations that IPOs could stimulate increased business registrations. In our analysis (results not presented), we observe that IPOs lead to additional small business lending and VC funding, which is in line with the notion that heightened entrepreneurial activity results in increased financing activity. We also acknowledge that these additional financing activities are also consistent with a “wealth effect” suggested by Butler et al. (2019), who show that local investors become wealthier due to IPOs and can fund new business ventures more actively.

for insiders. This shock may lead to an increased aggregate demand for local consumption goods and housing, subsequently boosting employment rates and house prices in the surrounding area (Butler, Fauver, and Spyridopoulos, 2019). This is primarily attributed to the wealth and liquidity shock generated by IPOs, which may result in increased local demand for consumer goods and housing. On the other hand, the effect of IPOs on aggregate growth may be negligible if the wealth shock triggered by IPOs remains localized and does not significantly impact aggregate demand in the local economy. In some cases, IPOs may even divert investment away from the local market, resulting in a negative effect on local economic growth (Cornaggia, Gustafson, Kotter, and Pisciotta, 2022).

While the parallel trends exhibited by our models suggest that our findings are not driven by such an effect, we further scrutinize this concern by looking directly for evidence of economic growth following a local IPO. In Table 6, we adopt a methodology similar to Barrios et al (2020) and estimate models using average weekly wages and employment as the outcomes. We employ a comparable specification to the previous analyses. We fail to document significant increases in average weekly wages or employment post IPO activity in the local area. The absence of such significant increases suggests that the surge in entrepreneurial activity following an IPO is unlikely to be solely attributed to the growth of the local economy. Our placebo test results would seem to suggest negligible immediate aggregate local effects of IPOs on local real labor market activities.⁶

⁶ Notably, the absence of a wage and employment effect can coexist with our findings, as the act of new business startups may lead to potential future growth with a lag and not necessarily at the time of the IPO as it takes years between a start-up and the business being an employer.

Our results provide valuable insights in the context of existing literature on the impact of IPOs on local economic growth. Importantly, prior research underscores that the relationship between IPOs and local economic growth hinges on factors such as the size of the IPO, with larger IPOs potentially having a more adverse impact on the local economy (Cornaggia, Gustafson, Kotter, and Pisciotta, 2022).

IV. INFORMATION MECHANISM ANALYSIS

In this section, we delve into the impact of the information generated by new IPO firm disclosures on the relationship between IPOs and new business formation. We explore how this information, by reducing uncertainty surrounding future returns from entrepreneurial activities, contributes to an amplified positive association between IPOs and new business registrations.

A. Economic Uncertainty

To assess the significance of IPOs' information production, we investigate a scenario where the firm's information should be particularly valuable to entrepreneurs. Specifically, we focus on locations characterized by higher levels of economic uncertainty, which, in turn, contribute to greater uncertainty in entrepreneurial income, especially before the IPO event.⁷ We expect that the role of the direct information produced by the IPO's disclosures will be most pronounced in counties where economic uncertainty is high. To measure economic uncertainty, we utilize a proxy based on the volatility in wage growth and employment growth within each

⁷ Conceptually, we can think of economic profits as reflecting demand shocks to industries, which in turn also lead to variation in wage growth. Under a rent-sharing perspective, whenever there is a demand shock that leads to change in profits, this change is shared between the firm and labor (Kline et al., 2019).

county (Barrios et al., 2021).⁸ We derive our measure from data in the Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW). To be precise, we calculate wage growth volatility as the combined sum of variances and covariances of wage growth rates across different industry sectors in the past 12 quarters. Employment growth volatility is calculated similarly. These calculations are weighted based on the employment share of each specific industry within the county. Mathematically, the measure of wage growth volatility for the portfolio of industries in a given county is expressed as:

$$\sigma_c^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} \sum_{j \neq i} w_i w_j \sigma_{ij} \quad (5)$$

where w_i denotes the proportion of total employment in industry i , σ_i^2 denotes the variance of wage growth rate in industry i , and c denotes county. Figure 4 shows the relationship between our wage growth volatility measure and new business registration. We absorb time and location fixed effects. As the scatter plot demonstrates, the relationship between wage growth volatility and entrepreneurial entry is negative, consistent with economic uncertainty discouraging entrepreneurial activity. Finally, we derive a principal component (PCA) score from wage growth volatility and employment growth volatility in a county-quarter.

In Table 7, we incorporate our proxy for economic uncertainty – *High Ex ante Uncertainty* is an indicator variable that equals 1 if the uncertainty PCA score is in the highest quartile across all counties in a specific quarter. These models include the

⁸ Ideally, we would use variation in economic profits in an area to proxy for economic uncertainty, but data on business profits is unavailable.

interaction of the *High Ex ante Uncertainty* indicator with the *POST IPO* variable. Supporting the notion of a direct information channel, the models in Table 7 indicates that the positive effects of the IPOs on new business formation are concentrated in counties with higher ex-ante economic uncertainty. Specifically, the estimates translate into an additional 30 percentage point increase in new business registrations for the top quartile of ex-ante economic uncertainty. This result reinforces the notion that new public firm disclosures provide valuable information to potential entrepreneurs.

Lastly, we strengthen the validity of our information mechanism hypothesis by examining a scenario where public firm information is likely to be of greater value to entrepreneurs. We specifically examine whether the effects we have identified are more pronounced for IPOs of new, independent, private firms compared to spinoff IPOs of public firm subsidiaries. In the case of subsidiary spinoffs, there is a reduced likelihood of generating significant additional information, as the parent corporation had already been providing public disclosures prior to the subsidiary's IPO. To delve into this comparison, we turn to Table 8, where we observe that the surge in entrepreneurial activity primarily occurs around IPOs of private firms. This observation aligns with our expectations, as these IPOs are associated with the release of plausible new information, making it more valuable to prospective entrepreneurs and, consequently, leading to increased entrepreneurial entry.

B. Information Acquisition

Our existing evidence reveals a substantial impact of local IPOs on new business formation, especially in situations characterized by heightened economic uncertainty, where the value of information is more pronounced. Nevertheless, this evidence does not fully elucidate the

specific nature of the information sought by entrepreneurs when making decisions about entering the market. For instance, IPOs can serve as signals to potential entrepreneurs, suggesting that engaging in business ventures could lead to success. While this “signaling effect” might be seen as conveying information regarding the likelihood of success, we propose the hypothesis that the information spillover from IPOs goes beyond mere signaling and also encompasses the direct production of information that entrepreneurs actively utilize in their decision-making.

An important ramification of a direct information mechanism contributing to the positive link between IPOs and new business formation is an expectation of intensified information acquisition patterns following the IPO event. In this section, we present evidence regarding the information acquisition behavior of local entrepreneurs or aspiring entrepreneurs subsequent to a local IPO, which include their engagements in downloading Edgar filings and the potential influence of the specific information content contained in the S-1 form filed by the IPO firm.

We initiate our examination by investigating the relation between IPOs and Edgar downloads. The results of this analysis are reported in Table 9. In Panel A, the dependent variable is the natural logarithm of the number of post-IPO Edgar downloads in the county quarter, and the independent variable of interest is an indicator for post-IPO in the county. The sample period spans the years 2005 through 2016. We utilize three different measures of Edgar downloads. In column (1), $\text{Log}(1 + \text{sum all})$ is defined as the log of one plus the total number of Edgar downloads across all file types. In columns (2) and (3), $\text{Log}(1 + \text{sum S1})$, and $\text{Log}(1 + \text{sum 10K})$ are defined as log of one plus the total number of downloads for S1 and 10K filings. Regardless of the specification of the dependent variable, we observe a strong positive and a statistically significant increase in Edgar downloads following an IPO in the county. In Panel

A2, we further confirm that the results continue to hold when we instrument *Post IPO* with the NASDAQ return IV.

In Panel B of Table 9, we regress *New Business Registrations* on indicators for the top quartile of each of our measures of Edgar downloads across counties in a year, as well as controls. Standard errors are robust and clustered at the county-level. Consistent with information from the IPO driving new business entry, we observe a positive and significant relationship between high downloads of S-1 filings and new business registrations. In contrast, the other types of filings do not load. Once again, the estimates are consistent with a role for information in driving new business formation in the county following a public offering of a local firm.

To focus on the role of the firm's newly disclosed information in explaining the relation between IPOs and new business starts, we go on to examine further the extent to which the downloading of public filings serves to magnify the relation between an IPO and new business formation. Specifically, in Panel C of Table 9, we regress our measure of *New Business Registrations* on interactions of the post-IPO indicator with high Edgar downloading indicators. Explicitly, we estimate the following equation:

$$\begin{aligned}
 & \text{Log}(1 + \text{New Business Reg})_{c,q} \\
 &= \delta \text{Post IPO}_{c,q} + \gamma_{10K} \text{Post IPO}_{c,q} * \text{High}(\text{sum}_{10K}) + \gamma_{S1} \text{Post IPO}_{c,q} \\
 & \quad * \text{High}(\text{sum}_{S1}) + \alpha_c + \gamma_q + \varepsilon_{c,q}
 \end{aligned}
 \tag{6}$$

Panel C of the table presents the estimates. We observe that the relationship between the IPO and new business formation is, in fact, magnified by the reliance on public filings, especially

the S-1s. This further bolsters the information channel as the mechanism for the positive relationship observed between an IPO and new business formation above. These findings are consistent with the view that public firm disclosures inform entrepreneurs, raising the bar for an alternative mechanism, such as industry growth prospects, to explain our results.

Overall Table 9 findings reveal that downloads of public firm filings from Edgar are higher in counties that recently experienced an IPO. Subsequently, we show that new business registrations are higher in counties that actively download these filings. Moreover, we find that the positive relation between an IPO and new business formation is amplified in counties with high levels of S-1 downloads. This evidence collectively highlights the significance of information acquisition as a driving force behind the increase in entrepreneurial activity following a local IPO.

We then refine our analysis by focusing on Edgar download activities of sophisticated entrepreneurs, presuming that sophisticated entrepreneurs are more inclined to use public disclosure information. Typically, innovation-driven new businesses are founded by more sophisticated entrepreneurs compared to new small businesses or traditional businesses such as corner grocery stores or laundries. Crunchbase tracks innovation-driven startups, and we identify the IP addresses of these firms using name matching, as described in Section II. We subsequently match these IP addresses to Edgar downloads to gauge the download activity of each startup. This approach allows us to zero in on the information acquisition and business activities of sophisticated entrepreneurs around IPOs. We develop two measures: the log number of Crunchbase firms that download at least one Edgar file in a county-year ($\text{Log}(1 + \text{Edgar Downloading Entrepreneurial Firms})$) and the log average number of Edgar file downloaded in a county-year ($\text{Log}(1 + \text{Entrepreneurial Downloads})$).

The estimates are presented in Table 10. We observe that both the number of Crunchbase firms downloading at least one Edgar filing in a county-year and the average number of Edgar files downloaded in a county-year are positively associated with the *Post IPO* variable. These positive relations suggest that Crunchbase entrepreneurs become more active in using public disclosure information as new IPO firms disclose information to the market. In economic terms, we document a 14% increase in Edgar downloading Crunchbase firms, and a 26% increase in total Edgar downloading activity among Crunchbase firms. To contextualize these findings, the documented effect implies an additional 4.6 Crunchbase firms downloading from Edgar (equivalent to 32 additional entrepreneurial Edgar downloads) per year, relative to the baseline average.⁹

Next, we delve into the S-1 forms submitted by the IPO firm to scrutinize the specific information content that potentially influences entrepreneurs' business formation decisions. Specifically, we undertake textual analysis to extract three dimensions of information content within these forms: the number of words, complexity as measured by the Fog Index, and the presence of highly pertinent information for prospective entrepreneurs. We interpret a higher number of words and greater language complexity as indicators of richer information content within S-1 forms. It is worth noting that previous studies have recognized that managerial linguistic complexity in financial filings and conference calls can be associated with either obfuscation or more informative technical disclosure (Bushee, Gow, and Taylor 2018). However, since managers' incentive to obfuscate information is likely to be low in S-1 forms,

⁹ Specifically, we multiply the 14% Crunch Base firm effect (26% total Edgar downloading effect) to their respective raw averages of 0.12 (0.45) and multiply that number to the average number of yearly IPOs during our sample, which is 272.

we believe that higher linguistic complexity is more likely to reflect a greater level of informative technical disclosure.

In addition to the conventional measures of information content such as the number of words and linguistic complexity frequently employed in prior literature, we introduce a novel entrepreneurial-focused metric called the "Entrepreneurship Score." This innovative measure is inspired by the work of Hassan et al. (2019). Specifically, to construct the Entrepreneurship Score, we identify keywords within the S-1 forms and in an Entrepreneurship textbook, and we calculate co-occurrences using a sliding window of words. This process results in a vector of keywords (bag-of-words) and the respective frequency of each keyword. We then compute this vector for each S-1 filing and the Entrepreneurship textbook, subsequently calculating the cosine similarity between the keyword frequency vector of each S-1 filing and the frequency vector of the Entrepreneurship textbook.

Our dataset comprises 1,326 S-1 filings from Edgar, successfully matched with respective IPO firms since 1996. Subsequently, we narrow our sample down to counties that have at least one IPO for which we possess textual information from the S-1 form. We form an indicator variable, *High S1 Text PCA Score*, that equals 1 if the PCA scores of the number of words, Fog index, and the Entrepreneurship Score are in the highest quartile across all S-1 forms. Subsequently, we employ a D.D. design, mirroring the approach in our primary regressions, to compare the impact of IPOs on subsequent new business formation between IPOs characterized by high information content in their S-1 forms and IPOs with lower values in these characteristics. Our findings, presented in Table 11, reveal that IPOs characterized by higher information content in S-1 forms exhibit a stronger association with subsequent new business formation. Economically, IPOs with high information content demonstrate an impact on new

business formation that is 4 to 15 times larger than IPOs with lower information content in their S-1.

C. Falsification Test

We next examine the extent to which the direct information channel drives our main findings by conducting a falsification test using new business registrations in the United Kingdom. This test is in the spirit to the one detailed in Badertscher et al. (2013). The premise behind this analysis is that an IPO should not exert a material impact on the local information environment within the UK context, as financial data for private companies is readily available in that setting. To conduct the exercise, we gathered data on the number of newly registered firms and information pertaining to IPO firms at the county-quarter equivalent level from the Orbis database. Subsequently, we executed a D.D. regression employing the same specifications as in our primary research design. The results presented in Online Appendix Table OA4 indicate that local IPOs do not exhibit any discernible effect on subsequent new business formation. This finding aligns with our hypothesis that IPOs likely influence new business formation primarily through an information channel, which is absent in the UK setting.

V. SURVEY-BASED EVIDENCE OF INFORMATION SPILLOVERS FROM IPOs

In this section, we complement our observational data-based evidence of information spillovers from IPOs with survey data collected from 503 individuals, encompassing both existing and prospective entrepreneurs. Our survey explores the impact of local IPO awareness on entrepreneurial intentions and preferences for information sources. Moreover, it provides further insight into the role of information spillovers from IPOs in influencing entrepreneurial activity.

We conducted a survey through the survey vendor Prolific to gather responses from a diverse sample of entrepreneurs. The survey included questions pertaining to the respondents' likelihood of initiating a new business when informed of a local IPO and their preferences for specific types of information related to IPOs, along with the sources from which they would obtain this information. Our sample included both individuals with and without prior entrepreneurial experience, and from various industries.¹⁰ Furthermore, over half of our survey participants possess a college degree or higher qualification. Responses are measured using a 5-point Likert Scale, ranging from 1 (Much Less Likely) to 5 (Much More Likely). The full survey instrument can be found in the online appendix.

Our survey yielded several key findings, which we report in Table 12. First, our survey respondents reported an awareness of an average of 2.07 IPOs within their respective industries over the past 5 years, with 1.54 of these IPOs occurring in their own regions. Upon learning about a local IPO, a significant majority of survey respondents exhibited either a neutral stance or an increased inclination to embark on a new entrepreneurial venture. Upon being informed about a local IPO, a significant majority of survey respondents exhibit either a neutral stance or an increased inclination to embark on a new entrepreneurial venture. The mean response is 3.31, which is statistically higher than the scale's midpoint (two-tailed, $p\text{-value} < 0.01$). Additionally, 35.58% of entrepreneurs responded with a value of 4 or greater.

Moving to the first question regarding the information spillover channel. Table 12, Panel B reports that nearly half of the respondents confirmed that newly available information from a

¹⁰ Our survey encompasses a diverse range of industries, including agriculture, construction, manufacturing, wholesale and retail trade, information technology, finance, real estate, transportation, professional services, and education.

local IPO positively influenced their decision to pursue a new entrepreneurial venture. The mean response is 3.57, which significantly exceeds the midpoint of the scale (two-tailed, p -value < 0.01). Moreover, 54.87% of participants providing a rating of 4 or higher. Interestingly, these figures are higher than those for the previous question, indicating that the information spillover derived from new IPO events holds greater value for entrepreneurs in their new business endeavors compared to the IPO events themselves. These findings highlight the importance of information spillovers in fostering entrepreneurial activity.

Finally, the survey revealed that entrepreneurs primarily seek insights and information from local IPOs concerning critical areas such as product market conditions, operational strategies, technologies, valuation, and potential funding opportunities. Entrepreneurs predominantly rely on traditional media, the IPO company's official website or SEC filings, and social media platforms to access the pertinent information from local IPOs.

Collectively, the survey results have significant implications for understanding the role of IPOs in influencing entrepreneurial intentions and the importance of information spillovers in the process. These findings provide an additional layer of evidence in support of the positive relationship between IPOs and new business formation. It is important to note, however, that while information spillover is a crucial factor, it may not necessarily be the exclusive mechanism linking local IPOs to subsequent business formation.

VI. CONCLUSION

In this paper, we delve into the impact of IPO activity within a local region on new business formation. The geographic distribution of entrepreneurial activity holds critical significance for policymakers striving to harness the economic growth generated by new businesses. Our

research demonstrates that local IPOs play a pivotal role in accelerating the establishment of new businesses within the county.

We explore a potential mechanism underlying this effect: information spillovers. Specifically, we investigate whether the public accounting disclosures of newly IPO'ed firms provide valuable information that informs entrepreneurs' decisions to initiate new businesses and guides financiers in their support of these enterprises. Our findings indicate that information emanating from public firm disclosures serves to reduce uncertainty related to entrepreneurial income volatility. This reduction in uncertainty acts as a catalyst, driving increased entrepreneurial activity via small business loans and venture capital investments. Moreover, it encourages potential entrepreneurs to participate in new business formation.

Our research addresses two fundamental questions, one in the realm of accounting and the other in the field of entrepreneurship. In the domain of accounting, a central inquiry pertains to the role of financial reporting in facilitating capital allocation to investment opportunities (Roychowdhury et al., 2019). Prior literature has primarily focused on the intensive margin—the enhancement of a firm's investment efficiency. Our paper builds upon this foundation, extending the inquiry to explore how public firm disclosures impact learning on the extensive margin, specifically in the context of new business formation.

Furthermore, our findings contribute to the fundamental question of what drives new business formation. Economists, dating back to Adam Smith, have underscored the significance of entrepreneurs and new business creation for the overall economy. Our results provide robust support for the existence of an information channel that significantly influences local entrepreneurial activity. This, in turn, calls for further exploration into the role of information and its accessibility in driving new business formation. We anticipate future research that will

continue to explore this emerging frontier at the intersection of information and entrepreneurial activities, both within the scope of this paper and in different settings.

APPENDIX A. VARIABLES DEFINITION

Name	Definition	Source
<i>Log(1+New Business Reg)</i>	Log of one plus the number of new business registrations	Startup Cartography
<i>Post IPO</i>	Indicator variable that is set to one in county quarters in the 3 years after an IPO event	Thomson SDC
<i>Log Population</i>	Log of total population in the area	Census
<i>Log Num Pub Firm</i>	Log of the number of public listed firms in the area, where area is either state or county	Compustat
<i>Log Income Per Capita</i>	Log of income per capita in the area	Census
<i>High Entr Search Share</i>	Indicator variable equal to one if entrepreneurial search is in the top quartile. Entrepreneurial search includes: “start a business,” “start your own business,” “start a company,” “how to incorporate,” “entrepreneurship,” “become an entrepreneur,” and “small business loan.”	Google Search
<i>Avg Wage Growth</i>	Average wage growth in a county-year	Census
<i>Avg Emp Growth</i>	Average employment growth in a county-year	Census
<i>High Ex-ante Uncertainty</i>	An indicator variable that equals 1 if uncertainty PCA score is in the highest quartile across all observations. The uncertainty PCA score is the PCA of two uncertainty variables, wage growth volatility, defined as the standard deviation of the past 12 quarters’ wage growth in a specific county, and employment growth volatility, defined as the standard deviation of the past 12 quarters’ employment growth in a specific county.	Census
<i>Log(1+Establishment Birth)</i>	Log of one plus the number <i>Establishment Births</i> . Where <i>Establishment Births</i> are of establishments that have zero employment in the first quarter of the initial year and positive employment in the first quarter of the subsequent year.	Census
<i>Income Growth</i>	Growth of income in the area	Census
<i>Log(1+sum all)</i>	Log of one plus the total number of Edgar downloads across all file types.	SEC
<i>Log(1+Edgar Downloading Entrepreneurial Firms) and Log(1+Entrepreneurial Downloads)</i>	Log(1+Edgar Downloading Entrepreneurial Firms) is the log number of CrunchBase firms that download at least one Edgar file in a county-year. Log(1+Entrepreneurial Downloads) is the log total number of Edgar files downloaded in a county-year.	CrunchBase
<i>High S1</i>	A dummy variable equal to 1 if the total number of downloads for S1 filings is in the highest quartile across counties in a year, and zero otherwise.	SEC
<i>High 10K</i>	A dummy variable equal to 1 if the total number of downloads for 10K filings is in the highest quartile across counties in a year, and zero otherwise.	SEC

High S1 Text PCA Score

An indicator that equals 1 if the PCA score of the number of words, Fog index, and the Entrepreneurship Score of the S-1 form of the IPO is in the highest quartile across all S-1 forms.

REFERENCES

- Admati, A. R. and P. Pfleiderer (2000). Forcing firms to talk: Financial disclosure regulation and externalities. *Review of Financial Studies* 13 (3), 479–519.
- Aghamolla, C., & Guttman, I. (2021). Strategic timing of IPOs and disclosure: A dynamic model of multiple firms. *The Accounting Review*, 96(3), 27–57.
- Aghamolla, C., & Thakor, R. T. (2022). IPO peer effects. *Journal of Financial Economics*, 144(1), 206–226.
- Armstrong, C. S., W. R. Guay, and J. P. Weber (2010). The role of information and financial reporting in corporate governance and debt contracting. *Journal of Accounting and Economics* 50 (2-3), 179–234.
- Babina, T., P. Ouimet, and R. Zarutskie. (2020). IPOs, Human Capital, and Labor Reallocation. Available at SSRN: <https://ssrn.com/abstract=2692845> or <http://dx.doi.org/10.2139/ssrn.2692845>
- Badertscher, B., N. Shroff, and H. D. White (2013). Externalities of public firm presence: Evidence from private firms' investment decisions. *Journal of Financial Economics* 109(3), 682–706.
- Baker, A. C., D. F. Larcker, and C. C. Y. Wang (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics* 144 (2), 370–395.
- Barrios, J. M. (2021). Staggeringly problematic: A primer on staggered DiD for accounting researchers. Available at SSRN 3794859.
- Barrios, J. M., Y. V. Hochberg, and H. Yi (2020). Launching with a parachute: The gig economy and entrepreneurial entry. *University of Chicago, Becker Friedman Institute for Economics Working Paper* (2020-21).
- Beatty, A., S. Liao, and J. J. Yu (2013). The spillover effect of fraudulent financial reporting on peer firms' investments. *Journal of Accounting and Economics* 55 (2- 3), 183–205.
- Berger, P. G. (2011). Challenges and opportunities in disclosure research—a discussion of ‘the financial reporting environment: Review of the recent literature’. *Journal of Accounting and Economics* 51 (1-2), 204–218.
- Bernard, D., Blackburne, T., & Thornock, J. (2020) Information flows among rivals and corporate investment. *Journal of Financial Economics*, 136(3), 760–779.
- Bernard, Darren, Nicole L. Cade, and Frank Hodge (2018). “Investor behavior and the benefits of direct stock ownership.” *Journal of Accounting Research* 56, no. 2: 431–466.
- Bernstein, S. (2015). Does Going Public Affect Innovation? *Journal of Finance*, 70(4), 1365–1403.
- Borisov, A., Ellul, A., & Sevilir, M. (2021). Access to public capital markets and employment growth. *Journal of Financial Economics*, 141(3), 896–918.

- Bound J, Jaeger D, Baker R. (1995). Problems with instrumental variables estimation when the correlation between the instruments and the endogeneous explanatory variable is weak. *J. Am. Stat. Assoc.* 90:443–50
- Breuer, Matthias, (2022), Bartik Instruments: An Applied Introduction, *Journal of Financial Reporting* 7, 49–67.
- Breuer, M. (2021). How Does Financial-Reporting Regulation Affect Market-Wide Resource Allocation? *Journal of Accounting Research* 59(1), 59-110.
- Breuer, M., C. Leuz, and S. Vanhaverbeke. (2020). Reporting Regulation and Corporate Innovation. Available at SSRN: <http://dx.doi.org/10.2139/ssrn.3449813>
- Bushee, B. J., I. D. Gow, and D. J. Taylor. 2018. Linguistic complexity in firm disclosures obfuscation or information? *Journal of Accounting Research* 56 (1): 85–121.
- Butler, A., Fauver, L., & Spyridopoulos, I. (2019). Local Economic Spillover Effects of Stock Market Listings. *Journal of Financial and Quantitative Analysis*, 54(3), 1025-1050.
- Call, Andrew C., Paul Hribar, Douglas J. Skinner, and David Volant. (2023) “Corporate managers’ perspectives on forward-looking guidance: Survey evidence.” Working Paper
- Celikyurt, U., Sevilir, M., & Shivdasani, A. (2010). Going public to acquire? The acquisition motive in IPOs. *Journal of Financial Economics*, 96(3), 345-363.
- Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The Effect of Minimum Wages on Low-Wage Jobs. *The Quarterly Journal of Economics* 134 (3), 1405–1454.
- Cornaggia, J., Gustafson, M., Kotter, J. D., & Pisciotta, K. (2022). Initial Public Offerings and the Local Economy. Available at SSRN https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3036176
- Davis, S. J., & Haltiwanger, J. (1992). Gross job creation, gross job destruction, and employment reallocation. *The Quarterly Journal of Economics*, 107(3), 819-863.
- Davis, S. J., Haltiwanger, J. C., & Schuh, S. (1998). Job creation and destruction. MIT Press Books.
- Decker, R., J. Haltiwanger, R. Jarmin, and J. Miranda (2014). The role of entrepreneurship in us job creation and economic dynamism. *Journal of Economic Perspectives* 28(3), 3–24.
- Decker, R. A., Haltiwanger, J., Jarmin, R. S., & Miranda, J. (2017). Declining dynamism, allocative efficiency, and the productivity slowdown. *American Economic Review*, 107(5), 322-326.
- Dougal, C., Parsons, C. A., & Titman, S. (2015). Urban vibrancy and corporate growth. *The Journal of Finance*, 70(1), 163-210.
- Durnev, A. and C. Mangen (2009). Corporate investments: Learning from restatements. *Journal of Accounting Research* 47 (3), 679–720.

Fairlie, R. W., Miranda, J., & Zolas, N. (2019). Measuring job creation, growth, and survival among the universe of start-ups in the United States using a combined start-up panel data set. *ILR Review*, 72(5), 1262-1277.

Guzman, J., & Stern, S. (2020). The state of American entrepreneurship: New estimates of the quantity and quality of entrepreneurship for 32 US States, 1988–2014. *American Economic Journal: Economic Policy*, 12(4), 212-243.

Hanlon, Michelle, and Nemit Shroff (2022). “Insights into auditor public oversight boards: Whether, how, and why they “work”.” *Journal of Accounting and Economics* 74, no.1: 101497.

Hassan, T. A., S. Hollander, L. van Lent, and A. Tahoun (2019): “Firm-Level Political Risk: Measurement and Effects,” *The Quarterly Journal of Economics*, 134, 2135–2202.

Haltiwanger, J., R. S. Jarmin, and J. Miranda (2013). Who creates jobs? Small versus large versus young. *Review of Economics and Statistics* 95 (2), 347–361.

Hathaway, I., & Litan, R. E. (2014). Declining business dynamism in the United States: A look at states and metros. Brookings institution.

Kim, J. and Valentine, K. (2021) The innovation consequences of mandatory patent disclosures. *Journal of Accounting and Economics*, 71(2-3), 1-22.

Kim, J., Valentine, K., (2022). Corporate financial disclosures and the market for innovation. Working paper, 1-65.

Kline, P., Petkova, N., Williams, H., & Zidar, O. (2019). Who profits from patents? rent-sharing at innovative firms. *The Quarterly Journal of Economics*, 134(3), 1343-1404.

Leonelli, Sinja, Maximilian Muhn, Thomas Rauter and Gurpal Sran. (2023) “How Do Consumers Use Firm Disclosure? Evidence from a Randomized Field Experiment” Working Paper

Leuz, C. and P. D. Wysocki (2016). The economics of disclosure and financial reporting regulation: Evidence and suggestions for future research. *Journal of Accounting Research* 54(2), 525–622.

Liberti, José María and Petersen, Mitchell A (2019). Information: Hard and Soft, *Review of Corporate Finance Studies*, Volume 8, Issue 1, March 2019, Pages 1–41,

Lowry, M., & Schwert, G. W. (2002). IPO market cycles: Bubbles or sequential learning? *Journal of Finance*, 57(3), 1171-1200.

Mitchell, M. L. and J. H. Mulherin. (1996). The impact of industry shocks on takeover and restructuring activity. *Journal of Financial Economics* 41, 193-229.

Roychowdhury, S., N. Shroff, and R. S. Verdi (2019). The effects of financial reporting and disclosure on corporate investment: A review. *Journal of Accounting and Economics* 68(2-3), 101246.

Ryans, James (2017), Using the EDGAR Log File Data Set. *Working Paper*, 1-36.

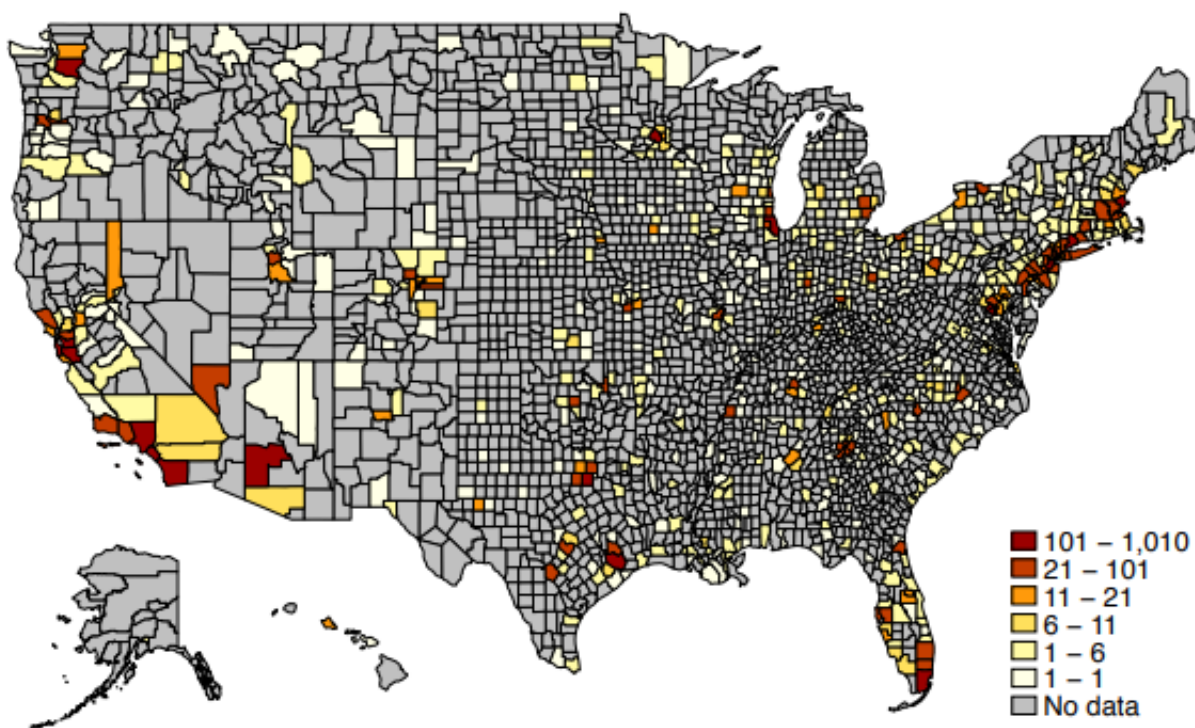
Staiger D, Stock J. 1997. Instrumental variables regression with weak instruments. *Econometrica* 65:557–86

Shroff, N., R. S. Verdi, and B. P. Yost (2017). When does the peer information environment matter? *Journal of Accounting and Economics* 64 (2-3), 183–214.

Shroff, N., R. S. Verdi, and G. Yu (2014). Information environment and the investment decisions of multinational corporations. *The Accounting Review* 89(2), 759–790.

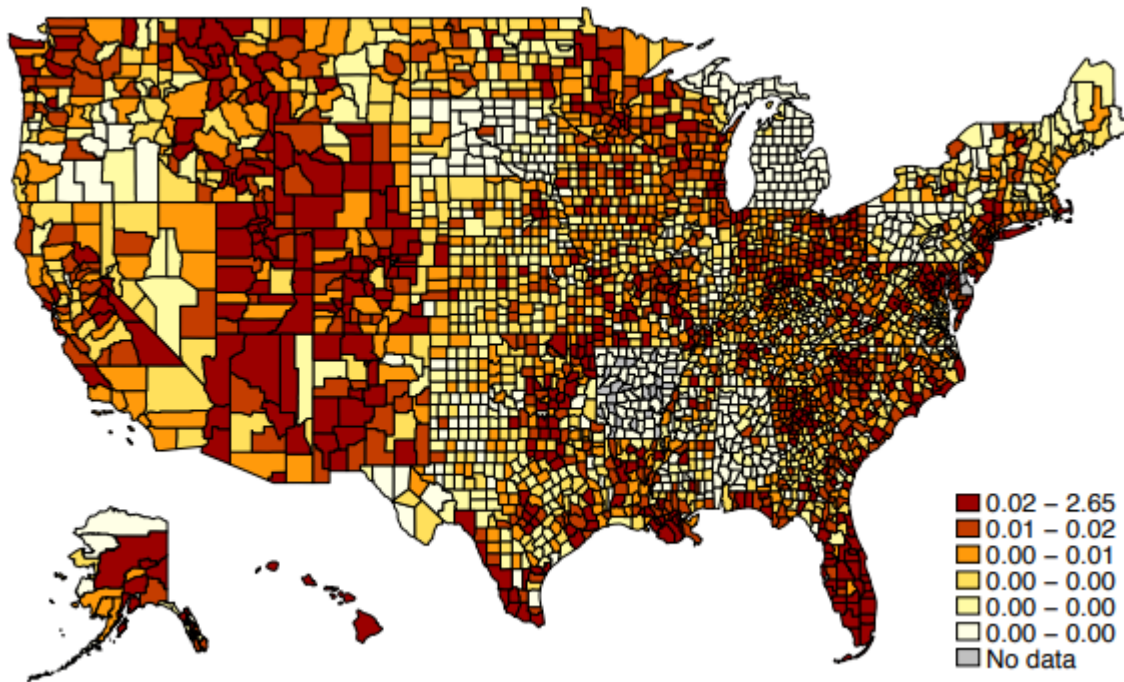
TABLES AND FIGURES

Figure 1. Distribution of IPOs by County



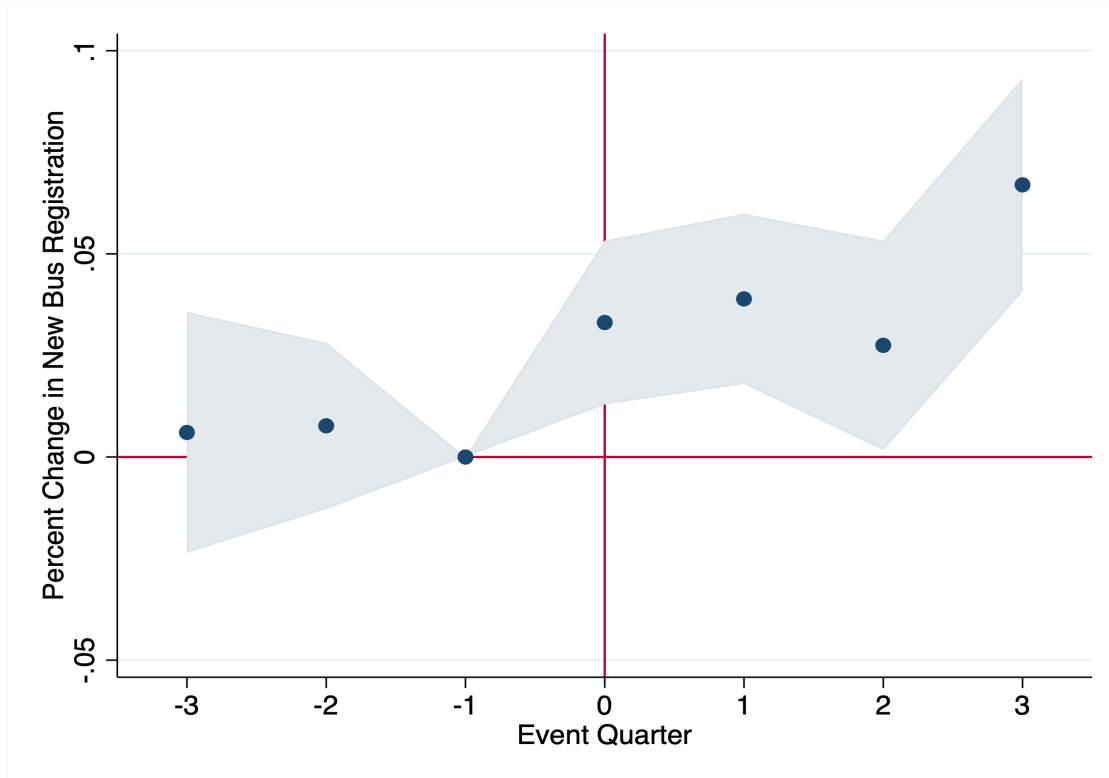
Notes: This figure plots the geographic distribution of 10,734 IPOs across U.S. Counties. The sample period is from 1985 Q1 to 2019 Q4.

Figure 2. New Business Registration by County



Notes: This figure plots the geographical distribution of annual average new business registration per capita across U.S. Counties. The sample period is from 1988 to 2016.

Figure 3. Parallel Trends of New Business Registration around an IPO Event



Notes: This figure presents the regression coefficient estimates in 7 quarters around the event window, from the regression in equation (1) with controls, time and location fixed effects. Each dot is a point estimate of δ in equation (1) by shifting *Post* from $Post_{q-3}$ to $Post_{q+3}$. The shaded region represents two-tailed 95% confidence intervals based on standard errors clustered at the county level. Detailed definitions of all variables are provided in Appendix A.

Figure 4. Economic Uncertainty and New Business Registration



Notes: The figure plots the relationship between the wage growth volatility (Ex-ante Uncertainty) and new business registration ($\text{Log}(1+\text{New Business Reg})$). This graph is based on a binned scatterplot partitioning the x-axis (i.e., the wage growth volatility) into 50 bins. The plots are based on averages conditional on the effects of our control variables (defined in Appendix A), as well as time and location fixed effects. Detailed definitions of all variables are provided in Appendix A.

Table 1. Descriptive Statistics

Panel A: Raw Sample

	(1) All				(2) County: At Least 1 IPO				(3) County: No IPO			
	Count	Mean	S.D.	P50	Count	Mean	S.D.	P50	Count	Mean	S.D.	P50
New Business Registration	359,892	371	2,451	18.00	75,800	1,313	4,626	174	284,092	119	1,265	11
Num Public Firm	359,892	0.10	1.06	0.00	75,800	0.46	2.28	0.00	284,092	0.00	0.07	0.00
Income Per Capita	359,892	26,670	11,181	24,808	75,800	31,518	134,745	29,341	284,092	25,376	10,098	23,758
Population	359,892	92,063	314,529	24,856	75,800	316,598	632,045	138,325	284,092	32,145	41,030	19,064
Observations	359,892				75,800				284,092			

Panel B: Stacked Sample

	All			
	Count	Mean	S.D.	P50
New Business Registration	474,759	1,606.64	5,030.52	53.00
Num Public Firm	474,759	1.91	7.48	0.00
Income Per Capita	474,759	29,285	13,011	26,666
Population	474,759	740,335	1,772,099	48,548
Observations	474,759			

Notes: Panel A presents descriptive statistics for the sample of new business registration data from the Startup Cartography Project. The sample is at the county-quarter level. Columns (2) and (3) show these statistics across counties with at least one IPO vs. counties with no IPO activity. Panel B presents descriptive statistics for the stacked sample used to estimate Model (1). In this sample, each IPO is treated as a separate event. Detailed definitions of all variables are provided in Appendix A.

Table 2. IPOs and New Business Registration

	(1) Log(1+New Bus Reg)	(2)	(3) Log(New Bus Reg)	(4)
Post IPO	0.039*** (2.78)	0.011*** (3.31)	0.029*** (3.14)	0.009*** (3.10)
Log Population	0.955*** (17.75)		0.993*** (20.82)	
Log Inc Per Capita	0.535*** (9.11)		0.660*** (12.94)	
Log # Pub Firm	0.112** (2.14)		0.044*** (3.25)	
Observations	474,759	474,759	418,856	418,856
Adj R-squared	0.889	0.946	0.919	0.950
Year-Quarter F.E.	Yes	No	Yes	No
County-Year F.E.	No	Yes	No	Yes
County F.E.	Yes	No	Yes	No
Quarter F.E.	No	Yes	No	Yes
Post Period	3 years	3 years	3 years	3 years
Cluster	County	County	County	County

Notes: This table reports estimation of Model (1) that regresses the number of new business registrations on IPOs. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Each IPO as a separate event (Complete Stack, 3-year window). Column (2) and Column (4) include the baseline regression with year-quarter and county fixed effects. Column (1) and (3) adds a vector of control variables. Column (3) and (4) exclude observations with zero new business registration. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 3. Causal Estimates of IPO on New Business Registration

Panel A. First Stage

	(1) IPO
NASDAQ Returns	0.307*** (6.48)
Observations	8,893
Adjusted R-squared	0.171
Controls	Yes
Year-Quarter F.E.	No
County-Year F.E.	Yes
County F.E.	No
Quarter F.E.	Yes
Cluster	County

Panel B. Fitted IPOs and New Bus Registration

	(1) Log(1+New Bus Reg)	(2) Log(New Bus Reg)
Post IPO Hat	0.014*** (3.35)	0.012*** (3.14)
Observations	474,759	418,856
Adjusted R-squared	0.946	0.950
Controls	Yes	Yes
Year-Quarter F.E.	No	No
County-Year F.E.	Yes	Yes
County F.E.	No	No
Quarter F.E.	Yes	Yes
Post Period	3 years	3 years
Cluster	County	County

Notes: Panel A presents the first-stage estimation of the instrumental variables analysis. Following Bernstein (2015), the dependent variable IPO is a dummy variable that equals to one if a firm completes an IPO filing, and zero otherwise. NASDAQ Returns is defined as two-month returns after the IPO filing date. Panel B presents the second-stage estimation using the fitted IPO values estimated from the first-stage in Panel A. Column (1) and (2) show the second stage estimation results include a vector of control variables with year-quarter and county fixed effects, while Column (2) excludes observations with zero new business registration. The unit of observation of Panel A is at the firm level while the unit of observation of Panel B is at the county quarter level. T-statistics, based on standard errors clustered at the County level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 4. IPOs and Establishment Births

Panel A		
	(1) Log(1+Establishment Birth)	(2) Log(Establishment Birth)
Post IPO	0.041*** (3.61)	0.030** (2.42)
Log Population	0.232*** (12.97)	0.353*** (12.52)
Log Income Per Capita	0.386*** (14.88)	0.500*** (14.96)
Log Num Pub Firm	0.011 (1.51)	0.014 (1.60)
Observations	788,242	529,600
Adjusted R-squared	0.740	0.743
Year F.E.	Yes	Yes
County F.E.	Yes	Yes
Industry	Yes	Yes
Post Period	3 years	3 years
Cluster	County	County
Panel B		
	(1) Log(1+Establishment Birth)	(2) Log(Establishment Birth)
Post IPO	0.033** (2.29)	0.028* (1.78)
Post IPO Other Industry	-0.012 (-0.95)	-0.002 (-0.17)
Log Population	0.232*** (12.97)	0.354*** (12.52)
Log Income Per Capita	0.385*** (14.87)	0.500*** (14.94)
Log Num Pub Firm	0.011 (1.54)	0.014 (1.60)
Observations	788,242	529,600
Adjusted R-squared	0.740	0.743
Year F.E.	Yes	Yes
County F.E.	Yes	Yes
Industry	Yes	Yes
Post Period	3 years	3 years
Cluster	County	County

Notes: Panel A reports estimation regressing the number of new establishment births Establishment Birth on IPOs. The unit of observation is at the county quarter industry level. Establishment Birth is the number of new establishments in an industry and county-year. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Column (1) and Column (2) include a vector of control variables with year, county and industry fixed effects, while Column (2) excludes observations with zero new business registration. Panel B reports estimation regressing the number of new establishment births Establishment Birth on IPOs and including Post IPO Other Industry in the regression. Post IPO Other Industry is an indicator variable equals one if IPOs in other industry exist in that county, and no IPO in the same industry. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A

Table 5. IPOs and Entrepreneurial Interest

Panel A		
	(1) P(High Entr Search Share)	(2) P(High Entr Search Share)
Post IPO	0.024** (2.43)	
Post IPO Hat		0.030** (2.43)
Observations	172,341	172,341
Adjusted R-squared	0.137	0.137
Controls	Yes	Yes
Year F.E.	Yes	Yes
State F.E.	Yes	Yes
Post Period	3 years	3 years
Cluster	County	County

Panel B		
	(1) Log(1+New Bus Reg)	(2) Log(New Bus Reg)
Post IPO	-0.050 (-0.50)	-0.146* (-1.71)
Post IPO X High Entr Search Share	0.387*** (3.11)	0.249** (2.09)
Observations	172,341	157,567
Adjusted R-squared	0.548	0.629
Controls	Yes	Yes
Year F.E.	Yes	Yes
State F.E.	Yes	Yes
Post Period	3 years	3 years
Cluster	County	County

Notes: Panel A reports estimation of Model (1) as a linear probability model that regresses entrepreneurial search intensity (P(High Entr Search Share)) on IPOs. P(High Entr Search Share) is an indicator variable that is defined as one in the case that the DMA is in the top quartile of entrepreneurial search intensity in that quarter. We measure entrepreneurial search intensity using internet searches for terms and phrases directly related to launching a business as in (Barrios et al., 2020). Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. The unit of observation is at the matched county (DMA) quarter level. Column (1) is the OLS model. Column (2) presents the second-stage estimation using the fitted IPO values estimated from the first-stage in Table 3 Panel A. Panel B shows the cross-sectional results of Post IPO * High Entr Search Share. Column (1) and Column (2) include a vector of control variables with year-quarter fixed effects. High Entr Search Share is an indicator variable that is defined as one in the case that the DMA is in the top quartile of entrepreneurial search intensity in that quarter. Column (2) exclude observations with zero new business registration. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 6. IPOs and Economic Growth

	(1) Avg Wage Growth	(2) Avg Emp Growth
Post IPO	-0.000** (-1.97)	-0.000 (-0.37)
Observations	233,391	233,391
Adjusted R-squared	0.483	0.237
Controls	Yes	Yes
Year-Quarter F.E.	Yes	Yes
County-Year F.E.	No	No
County F.E.	Yes	Yes
Quarter F.E.	No	No
Post Period	3 years	3 years
Cluster	County	County

Notes: This table reports estimation of regressions of average wage growth (Avg Wage Growth) and average employment growth (Avg Emp Growth) on IPOs. The sample period is from 2000 Q1 to 2016 Q4 as average wage growth and average employment growth data is available in this period. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Column (1) and Column (2) include a vector of control variables with year-quarter and county fixed effects. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 7. IPOs, New Business Registrations, and Uncertainty

	(1) Log(1+New Bus Reg)	(2) Log(New Bus Reg)
Post IPO X High Ex-ante Uncertainty	0.299** (2.12)	0.292** (2.17)
Post IPO	-0.181** (-2.13)	-0.240*** (-3.00)
Observations	91,099	82,890
Adjusted R-squared	0.906	0.936
Controls	Yes	Yes
Year-Quarter F.E.	Yes	Yes
County-Year F.E.	No	No
County F.E.	Yes	Yes
Quarter F.E.	No	No
Post Period	3 years	3 years
Cluster	County	County

Notes: This table reports estimation of how the impact of IPO on new business registrations depends on local economic uncertainty. The sample period is from 2000 Q1 to 2016 Q4 as wage growth data is available in this period. High Ex-ante Uncertainty is an indicator variable that equals 1 if uncertainty PCA score is in the highest quartile across all observations. The uncertainty PCA score is calculated based on two uncertainty variables. The first uncertainty variable is wage growth volatility, defined as the standard deviation of the past 12 quarters' wage growth in a specific county. The second variable is employment growth volatility, defined as the standard deviation of the past 12 quarters' employment growth in a specific county. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Column (1) and Column (2) include a vector of control variables with year-quarter and county fixed effects, while Column (2) excludes observations with zero new business registration. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 8. Spinoffs vs. Private to Public IPOs

	(1) Log(1+New Bus Reg)	(2) Log(1+New Bus Reg)
Post Spinoff IPO	0.014 (0.24)	
Post Private IPO		0.068** (2.04)
Observations	474,759	474,759
Control	Yes	Yes
Year-Quarter F.E.	Yes	Yes
County F.E.	Yes	Yes
Cluster	County	County
Adjusted R-squared	0.889	0.889

Notes: This table reports estimation of Model (1) that regress new business registrations on two types of IPOs. Post Public IPO is an indicator variable that equals one during the 3-year period after a subsidiary of an already public firm goes public, and zero if during the 3-year period before a subsidiary of an already public firm goes public. In contrast, Post Private IPO is an indicator variable that equals one during the 3-year period after a private firm goes public, and zero if during the 3-year period before a subsidiary of a private firm goes public. Each public/private IPO as a separate event (Complete Stack, 3-year window), post public/private IPO equals one if it is after the public/private IPO event. Control variables include the following three variables. Log Income Per Capita is the log of income per capita at the county-year level. Log Population is the log number of population at the county-year level. Log Num Pub Firm is the log number of public listed firms in the state. Standard errors clustered at the state level are reported in the parentheses. ***, **, and * mark statistical significance at 1%, 5%, and 10%, respectively.

Table 9. Edgar Searches and New Business Registrations

Panel A. Edgar Downloads on Post IPO

	(1) Log(1+Sum All)	(2) Log(1+Sum S1)	(3) Log(1+Sum 10K)
Post IPO	0.305*** (8.68)	0.351*** (9.27)	0.332*** (10.53)
Observations	158,895	158,895	158,895
Adjusted R-squared	0.975	0.961	0.932
Controls	Yes	Yes	Yes
Year-Quarter F.E.	Yes	Yes	Yes
County-Year F.E.	No	No	No
County F.E.	Yes	Yes	Yes
Quarter F.E.	No	No	No
Post Period	3 years	3 years	3 years
Cluster	County	County	County

Panel A2 - IV Second Stage

	(1) Log(1+Sum All)	(2) Log(1+Sum S1)	(3) Log(1+Sum 10K)
Post IPO Hat	0.373*** (8.60)	0.407*** (10.50)	0.428*** (9.15)
Observations	158,827	158,827	158,827
Adjusted R-squared	0.975	0.932	0.961
Controls	Yes	Yes	Yes
Year-Quarter F.E.	Yes	Yes	Yes
County-Year F.E.	No	No	No
County F.E.	Yes	Yes	Yes
Quarter F.E.	No	No	No
Post Period	3 years	3 years	3 years
Cluster	County	County	County

Panel B. New Business Registration in Counties with High Download

	(1) Log(1+New Bus Reg)	(2) Log(1+ New Bus Reg)	(3) Log(New Bus Reg)	(4) Log(New Bus Reg)
High S1	0.501*** (5.37)	0.478*** (5.09)	0.289*** (3.60)	0.295*** (3.55)
High 10K	-0.516*** (-5.93)	-0.508*** (-5.84)	-0.576*** (-7.25)	-0.566*** (-7.01)
Observations	158,895	158,895	145,104	145,104
Adj R-squared	0.506	0.509	0.583	0.585
Controls	Yes	Yes	Yes	Yes
Yr-Qtr F.E.	No	Yes	No	Yes
Cty-Yr F.E.	No	No	No	No
County F.E.	No	No	No	No
Quarter F.E.	No	No	No	No
Post Period	3 years	3 years	3 years	3 years
Cluster	County	County	County	County

Panel C. New Business Registration on Post IPO in counties with High Download

	(1) Log(1+New Bus Reg)	(2) Log(New Bus Reg)
Post IPO	-0.127*** (-3.05)	-0.149*** (-4.40)
Post IPO X High S1	0.139*** (2.69)	0.162*** (3.32)
Post IPO X High 10K	0.021 (0.47)	0.041 (0.99)
Observations	158,895	145,077
Adj R-squared	0.911	0.937
Controls	Yes	Yes
Year-Quarter F.E.	Yes	Yes
County-Year F.E.	No	No
County F.E.	No	No
Quarter F.E.	No	No
Post Period	3 years	3 years
Cluster	County	County

Notes: Panel A presents results from regressing the number of Edgar downloads of different filings on Post IPO (OLS and IV). The sample period is from 2005 to 2016. In Column (1) Log(1+Sum All) is the log of one plus the total number of Edgar downloads across all file types. In Columns (2) and Column (3), Log(1+sum S1) and Log(1+sum 10K) are defined as log of one plus the total number of downloads across S1 and 10K filings. In Panel B, High S1 and High 10K represent dummy variables equal to 1 if the total number of downloads for each respective filing type is in the highest quartile in a year, and zero otherwise. Column (1) and Column (3) of Panel B includes a vector of control variables while Column (2) and Column (4) adds the year-quarter fixed effect. Panel C shows the cross-sectional results of Post IPO * High S1 and Post IPO * High 10K. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 10. IPOs and Entrepreneurs' Edgar Searches

	(1) Log(1+Edgar Downloading Firms)	(2) Log(1+Entrepreneurial Downloads)
Post IPO	0.136*** (3.86)	0.262** (2.13)
Observations	9,732	9,732
Adjusted R-squared	0.847	0.807
Controls	Yes	Yes
Year F.E.	Yes	Yes
County-Year F.E.	No	No
County F.E.	Yes	Yes
Quarter F.E.	No	No
Post Period	3 years	3 years
Cluster	County	County

Notes: This table reports the results of the regression of entrepreneurial Edgar downloading behavior on IPOs. The sample is the universe of US based CrunchBase companies. The dependent variable in Column (1) Log(1+Edgar Downloading Firms) is defined as the log number of CrunchBase firms that download at least one Edgar file in a county-year. Log(1+Entrepreneurial Downloads) in Column (2) is defined as the log total number of Edgar file downloaded in a county-year. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Column (1) and (2) show the second stage estimation results include a vector of control variables with year-quarter and county fixed effects, while Column (2) excludes observations with zero new business registration. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the County level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table 11. S-1 Form Information Content and New Business Registrations

	(1) Log(1+New Bus Reg)	(2) Log(New Bus Reg)
Post IPO	0.004 (0.41)	0.011 (1.42)
Post IPO X High S1 text PCA Score	0.063*** (2.75)	0.043** (2.31)
Observations	70,378	70,101
Adjusted R-squared	0.946	0.978
Controls	Yes	Yes
Year-Quarter F.E.	Yes	Yes
County-Year F.E.	No	No
County F.E.	Yes	Yes
Quarter F.E.	No	No
Post Period	3 years	3 years
Cluster	County	County

Notes: This table reports the interaction effect of IPO and S-1 form information content on new business formation. High S1 Text PCA Score is an indicator that equals 1 if the PCA score of the number of words, Fog index, and the Entrepreneurship Score of the S-1 form of the IPO is in the highest quartile across all S-1 forms. Both column (1) and Column (2) include a vector of control variables with year-quarter and county fixed effects; Column (2) excludes observations with zero new business registration. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix.

Table 12. Entrepreneurship and IPO Survey Results

Panel A: How More and Less Likely Would You Start a New Business After Observing an IPO Event in Your Region?

Sample	Count (and %) of Responses					Total
	1 Much Less Likely	2	3	4	5 Much More Likely	
All	2	33	289	165	14	503
Respondents	0.40%	6.56%	57.46%	32.80%	2.78%	100%

Panel B: How More or Less Likely Would the Information Obtained from an IPO Event Affect Your Entrepreneurship Decision?

Sample	Count (and %) of Responses					Total
	1 Extremely Less Likely	2	3	4	5 Extremely More Likely	
All	1	22	204	241	35	503
Respondents	0.20%	4.37%	40.56%	47.91%	6.96%	100%

Panel C: What Type(s) of Information Do Entrepreneurs Learn from IPO Events?

Option	Proportion that Chose
Valuation	60.04% (N=302)
Product market (e.g., size and price)	54.87% (N=276)
Funding opportunities	47.51% (N=239)
Innovations and technologies	44.53% (N=224)
Operational strategies (e.g., production, distribution, supply chain)	39.96% (N=201)
Local investment opportunities	36.58% (N=184)
Partnership opportunities	34.39% (N=173)
Regulatory compliance	24.65% (N=124)
Other (please specify)	1.59% (N=8)

Panel D: What Source(s) Do Entrepreneurs Use Obtain This Information?

Option	Proportion that Chose
Traditional media (e.g., WSJ, Financial Times, NYT, CNN, Bloomberg)	62.43% (N=314)
Official sites (e.g., company's websites, sec.gov)	56.46% (N=284)
Social media (e.g., Twitter, LinkedIn, Facebook)	44.33% (N=223)
Networking (e.g., friends, colleagues, and peers)	43.74% (N=220)
Industry association and advisory service (e.g., consulting company, survey)	31.61% (N=159)
Other (please specify)	2.58% (N=13)

Note: This table presents the results from our survey about entrepreneurship and IPOs. Panel A presents the count of participant responses (and % of the total sample) in response to our survey question about how more or less likely an entrepreneur would start a new business after observing an IPO event in your region (from 1=Much Less Likely to 5 = Much More Likely). Panel B presents the count of participant responses (and % of the total sample) in response to our survey question about how more or less likely the information obtained from IPO events would affect an entrepreneurship decision (from 1=Extremely Less Likely to 5 = Extremely More Likely). All reported p-values regarding this table are two-tailed. Panel C of this table presents what type(s) of information in IPO events are useful for entrepreneurs. Panel D of this table presents what source(s) entrepreneurs use to obtain information about IPO events. All the panels in this table uses 503 survey participants' responses. Participants were able to select more than one type or source in their responses in Panels C and D. Internet Appendix contains the survey instrument.

ONLINE APPENDIX

Informing Entrepreneurs: Public Corporate Disclosure and New Business Formation

OA Table 1. Spillover Effects of IPOs on New Business Registration

OA Table 2. IPO and New Business Registration (Entropy Balancing)

OA Table 3. Public Firm Presence and Establishment Births

OA Table 4. Falsification Test with UK Setting

OA Table 5. Descriptive Statistics of Information Acquisition and Entrepreneur Financing

OA Table 6. Robustness Tests of Alternative Research Design

OA Sample Notes

OA Survey Instruments

Table OA1. Spillover Effects of IPOs on New Business Registration

	(1) Log(1+New Bus Reg)	(2) Log(1+New Bus Reg)	(3) Log(1+New Bus Reg)
Post IPO	0.060*** (4.32)		
NCounty Post IPO		0.038*** (2.50)	
2NCounty Post IPO			0.003 (0.26)
Observations	316,222	284,117	125,288
Controls	Yes	Yes	Yes
Year-Quarter F.E.	Yes	Yes	Yes
County F.E.	Yes	Yes	Yes
Cluster	County	County	County
Adjusted R-squared	0.914	0.849	0.839

Notes: This table reports estimation of Model (1) that regresses the number of new business registrations on IPOs. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Each IPO as a separate event (Complete Stack, 3-year window), Post-IPO equals one if it is after the IPO event. Columns (1) and (2) report the results of regressing new business registrations on Post IPO, excluding the IPO county's neighboring counties (Column (1)) and the IPO county but with the neighboring county (Column (2)), respectively. Column (3) excludes both the IPO neighboring counties and IPO counties. Specifically, NCounty Post IPO is an indicator variable that equals one during the 3-year period after a neighboring county has an IPO, and zero if during the 3-year period before a neighboring county has an IPO. 2NCounty Post IPO is an indicator variable that equals one during the 3-year period after a "second-degree" neighboring county has an IPO, and zero if during the 3-year period before a "second-degree" neighboring county has an IPO. Control variables include the following three variables. Log Income Per Capita is the log of income per capita at the county-year level. Log Population is the log number of population at the county-year level. Log Num Pub Firm is the log number of public listed firms in the state. Standard errors clustered at the county level are reported in the parentheses. ***, **, and * mark statistical significance at 1%, 5%, and 10%, respectively.

Table OA2. IPO and New Business Registration (Entropy Balancing)

	(1) Log(1+New Bus Reg)	(2) Log(1+New Bus Reg)	(3) Log(New Bus Reg)	(4) Log(New Bus Reg)
Post IPO	0.097*** (5.64)	0.114*** (6.53)	0.143*** (9.03)	0.129*** (7.41)
Log Population	0.900*** (6.79)		0.909*** (7.98)	
Log Income Per Capita	0.892*** (6.09)		0.929*** (6.82)	
Log Num Pub Firm	0.036 (0.73)		0.000 (0.02)	
Observations	474,759	474,759	418,856	418,856
Adjusted R-squared	0.897	0.971	0.937	0.973
Year-Quarter F.E.	Yes	No	Yes	No
County-Year F.E.	No	Yes	No	Yes
County F.E.	Yes	No	Yes	No
Quarter F.E.	No	Yes	No	Yes
Post Period	3 years	3 years	3 years	3 years
Cluster	County	County	County	County

Notes: This table reports estimation of Model (1) using entropy balanced sample that regresses the number of new business registrations on IPOs. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Column (2) and Column (4) include the baseline regression with year-quarter and county fixed effects. Column (1) and Column (3) adds a vector of control variables. Specifically, Column (3) and (4) exclude observations with zero new business registration; The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table OA3. Public Firm Presence and Establishment Births

	(1) Log(1+Establishment Birth)	(2) Log(Establishment Birth)
Pub Firm Presence Quintile	0.0890*** (0.0096)	0.1030*** (0.0157)
Observations	782,850	552,868
Controls	Yes	Yes
Year F.E.	Yes	Yes
County F.E.	Yes	Yes
Industry F.E.	Yes	Yes
Cluster	County	County
Adjusted R-squared	0.784	0.795

Notes: This table reports the estimation results of a new model that regresses the number of new establishment births at the county, industry, year level on the quintile rank of public firm presence in the industry. Establishment Birth is the number of new establishments in an industry and county-year. Pub Firm Presence Quintile is the quintile rank of public firm presence taken from Shroff et al. (2017) and defined as the number of public firms in an industry (3 digit SIC) divided by the total number firms in the industry, both public and private. Log Income Per Capita is the log of income per capita at the county-year level. Log Population is the number of population at the county-year level. Standard errors clustered at the county level are reported in the parentheses. ***, **, and * mark statistical significance at 1%, 5%, and 10%, respectively.

Table OA4. Falsification Test with UK Setting

	(1) Log(1+New Bus Reg) UK	(2) Log(1+New Bus Reg) UK	(3) Log(New Bus Reg) UK	(4) Log(New Bus Reg) UK
Post IPO	0.003 (0.32)	-0.001 (-0.45)	0.009 (0.90)	-0.001 (-0.38)
Log Population	-1.096 (-1.20)		-0.600 (-0.90)	
Log Income Per Capita	0.675 (1.49)		0.598 (1.56)	
Log Num Pub Firm	0.002 (0.05)		-0.045 (-1.56)	
Observations	31,784	31,784	31,712	31,712
Adjusted R- squared	0.987	0.993	0.986	0.992
Year-Quarter F.E.	Yes	No	Yes	No
County-Year F.E.	No	Yes	No	Yes
County F.E.	Yes	No	Yes	No
Quarter F.E.	No	Yes	No	Yes
Post Period	3 years	3 years	3 years	3 years
Cluster	County	County	County	County

Notes: This table reports estimation of Model (1) that regresses the number of new business registrations on IPOs, while changing the sample to UK. The New Bus Reg UK is defined as the number of firms incorporated in the Orbis database. Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Column (2) and Column (4) include the baseline regression with year-quarter and county fixed effects. Column (1) and (3) adds a vector of control variables. Specifically, Column (3) and (4) exclude observations with zero new business registration. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table OA5. Descriptive Statistics of Information Acquisition and Entrepreneur Financing

Panel A. Information Acquisition

	Obs.	Mean	SD	Median
All	49,091	660.33	5606.90	6.00
S1	49,091	8.48	65.57	0.00
10K	49,091	124.98	869.60	0.00

Panel B: Entrepreneurial Information Acquisition

	Obs.	Mean	SD	Median
Log(1+Edgar Downloading Entrepreneurial Firms)	9,732	0.12	0.39	0
Log(1+Entrepreneurial Downloads)	9,732	0.45	1.53	0

Notes: This table presents summary statistics for the sample of Edgar downloads (Panel A) from 2005 to 2016, and entrepreneurial Edgar downloads (Panel B) from 2005 to 2016. The sample is at the county-year level. In Panel A, All is defined as the total number of Edgar downloads across all file types, including 10K, 10Q, 8K, DEF 14A, S1 and others. S1, 10K, 10Q and Others are respectively defined as the total number of downloads of S1, 10K, 10Q and other filings (8K and DEF 14A).

Table OA6. Robustness Tests of Alternative Research Design

	(1)	(2)	(3)	(4)	(5)	(6)
	Log(1+New Bus Reg)		Log(New Bus Reg)		New Bus Reg (Poisson)	
Post IPO	0.048** (2.48)	0.048*** (3.33)	0.020 (1.51)	0.036*** (2.68)	0.552*** (6.03)	0.758*** (4.71)
Log Population	0.934*** (16.62)		0.993*** (19.75)		0.714*** (7.06)	
Log Inc Per Capita	0.524*** (8.83)		0.649*** (12.59)		0.781*** (4.10)	
Log # Pub Firm	0.057** (2.00)		0.017 (0.87)		- 0.063*** (-2.61)	
Observations	300,389	300,389	249,111	249,111	279,357	279,357
Adj R-squared	0.870	0.926	0.899	0.929	0.947	0.974
Year-Quarter F.E.	Yes	No	Yes	No	Yes	No
County-Year F.E.	No	Yes	No	Yes	No	Yes
County F.E.	Yes	No	Yes	No	Yes	No
Quarter F.E.	No	Yes	No	Yes	No	Yes
Post Period	3 years	3 years	3 years	3 years	3 years	3 years
Cluster	County	County	County	County	County	County

Notes: This table reports estimation of Model (1) that regresses the number of new business registrations on IPOs. In contrast to Table 2, we limit sample to only those with either no or only one IPO within 6 year period of a county-quarter (i.e., no overlapping with the other IPOs). Post IPO is an indicator variable that equals one during the 3-year period after an IPO, and zero if during the 3-year period before an IPO. Each IPO as a separate event (Complete Stack, 3-year window). Column (2) and Column (4) include the baseline regression with year-quarter and county fixed effects. Column (1) and (3) adds a vector of control variables. Column (3) and (4) exclude observations with zero new business registration. The unit of observation is at the county quarter level. T-statistics, based on standard errors clustered at the county level, are reported below the coefficient estimates in parenthesis. ***, **, * indicates statistical significance at 1%, 5%, and 10%, respectively (two-tailed). Detailed definitions of all variables are provided in Appendix A.

Table OA7: Table Sample and Specification Notes

Tables	Sample Size	Notes	Year-Quarter F.E.	County-Year F.E.	County F.E.	Quarter F.E.	Year F.E.	State F.E.	Industry F.E.
Table 2. IPOs and New Business Registration									
Column 1	✓ 474,759	Full Sample	Yes	No	Yes	No			
Column 2	✓ 474,759	Full Sample	No	Yes	No	Yes			
Column 3	✓ 418,856	Exclude New Firm Num = 0	Yes	No	Yes	No			
Column 4	✓ 418,856	Exclude New Firm Num = 0	No	Yes	No	Yes			
Table 3. Causal Estimates of IPO on New Business Registration									
Panel A - First Stage									
Column 1	8,893	First-Stage Sample	Yes	No	Yes	No			
Panel B - Fitted IPOs and New Business Registration									
Column 1	✓ 474,759	Full Sample	Yes	No	Yes	No			
Column 2	✓ 418,856	Exclude New Firm Num = 0	Yes	No	Yes	No			
Table 4. IPOs and Establishment Births									
Panel A									
Column 1	788,242	Industry-Year-County Sample			Yes		Yes		Yes
Column 2	529,600	Industry-Year-County Sample Exclude New Birth = 0			Yes		Yes		Yes
Panel B									
Column 1	788,242	Industry-Year-County Sample			Yes		Yes		Yes
Column 2	529,600	Industry-Year-County Sample Exclude New Birth = 0			Yes		Yes		Yes
Table 5. IPOs and Entrepreneurial Interest									
Column 1	172,341	Nielsen Designated Market Areas (DMAs) Matched Sample					Yes		Yes
Column 2	172,341	Nielsen Designated Market Areas (DMAs) Matched Sample					Yes		Yes
Table 6. IPOs and Economic Growth Placebo									
Column 1	✓ 233,391	Wage and Employment Sample	Yes	No	Yes	No			
Column 2	✓ 233,391	Wage and Employment Sample	Yes	No	Yes	No			
Table 7. IPOs, New Business Registrations, and Uncertainty									
Column 1	91,099	Wage Growth Volatility Sample (Start from 2007)	Yes	No	Yes	No			
Column 2	82,890	Wage Growth Volatility Sample (Start from 2007) Exclude New Firm Num = 0	Yes	No	Yes	No			
Table 8. Spilloffs vs. Private to Public IPOs									
Column 1	✓ 474,759	Full Sample	Yes	No	Yes	No			
Column 2	✓ 474,759	Full Sample	Yes	No	Yes	No			
Table 9. Edgar Searches and New Business Registrations									
Panel A. Edgar Downloads on Post IPO									
Column 1	✓ 158,895	Edgar Sample	Yes	No	Yes	No			
Column 2	✓ 158,895	Edgar Sample	Yes	No	Yes	No			
Column 3	✓ 158,895	Edgar Sample	Yes	No	Yes	No			
Panel A2. IV Second Stage									
Column 1	✓ 158,827	Edgar Sample with IV	Yes	No	Yes	No			
Column 2	✓ 158,827	Edgar Sample with IV	Yes	No	Yes	No			
Column 3	✓ 158,827	Edgar Sample with IV	Yes	No	Yes	No			
Panel B. New Business Registration in Counties with High S1/AQ/AQ Download									
Column 1	✓ 158,895	Edgar Sample	No	No	No	No			
Column 2	✓ 158,895	Edgar Sample	Yes	No	No	No			
Column 3	145,104	Edgar Sample Exclude New Firm Num = 0							
Column 4	145,104	Edgar Sample Exclude New Firm Num = 0							
Panel C. New Business Registration on Post IPO in counties with High S1/AQ/AQ downloads									
Column 1	✓ 158,895	Edgar Sample	Yes	No	No	No			
Column 2	✓ 145,077	Edgar Sample Exclude New Firm Num = 0	Yes	No	No	No			
Table 10. IPOs and Entrepreneurs' Edgar Searches									
Column 1	✓ 9,732	Edgar Download Firm Sample	Yes	No	Yes	No			
Column 2	✓ 9,732	Edgar Download Firm Sample	Yes	No	Yes	No			
Table 11. S1 Characteristics and New Business Registrations									
Column 1	70,378	S1 characteristics Sample	Yes	No	Yes	No			
Column 2	70,101	S1 characteristics Sample Exclude New Firm Num = 0	Yes	No	Yes	No			

Survey Instruments

Q1. What impact would the news of a private company going public through an IPO have on your business decisions as an entrepreneur? Enter text below (min. 20 characters).

e.g. "When a private company goes public through an IPO, i'd assume that means that financial records and other company information that used to be "behind the veil" are now a matter of public record. I would analyze the finances of the business, look into potentially unknown partnerships, etc."

Q2. As an entrepreneur, what of the following areas do you gain insights or information on when another company goes public through an IPO? Select all that apply.

- ☐ Product market (e.g., size and price)
- ☐ Operational strategies (e.g., production, distribution, supply chain)
- ☐ Innovations and technologies
- ☐ Valuation
- ☐ Funding opportunities
- ☐ Partnership opportunities
- ☐ Local investment opportunities
- ☐ Regulatory compliance
- ☐ Other (please specify)

Q3. Which information source(s) do you rely on most to acquire information about IPOs? Select all that apply.

- ☐ Traditional media (e.g., WSJ, Financial Times, NYT, CNN, Bloomberg)
- ☐ Social media (e.g., Twitter, LinkedIn, Facebook)
- ☐ Industry association and advisory service (e.g., consulting company, survey)
- ☐ Official sites (e.g., company's websites, sec.gov)
- ☐ Networking (e.g., friends, colleagues, and peers)
- ☐ Other (please specify)

Q4. If you were to learn that a private company in your local area (e.g., metro or county) is going public through an IPO, how would that affect your inclination to launch a new business? Would you be more or less likely to pursue a new venture?

Q5. When a company goes public via IPO, previously unavailable information may become available from multiple information sources (e.g., media and official websites). How much would the potential insights or information gained from these sources affect your decision to pursue a new venture?

Q6. Please elaborate on your answer to the question above: why would you be more or less likely to pursue a new venture given IPO related information?

e.g., "Depending on the market, market location, and tgr product or service offered, it could either boost my ideas or efforts to go public or it could deter my rationale. It's responsible to look at both the good and the bad of a new IPO as it relates to my ideas or ideal. Sometimes others experience help me out a lot to determine to move forward or step back. "

Survey Participant Demographics

Panel A: Entrepreneurial Experience

	No experience	less than 1 year	1-3 years	4-5 years	6-10 years	10+ years	Total
Respondents	133	51	138	81	53	47	503
	26.44%	10.14%	27.44%	16.10%	10.54%	9.34%	100%

Panel B: IPO Awareness in Past 5 Years

	Total IPOs	Industry-specific IPOs	Local IPOs
Mean	10.88	1.807	1.296

Panel C: Education

	Less than high school degree	High school graduate	Some college	Associate degree in college (2- year)	Bachelor's degree in college (4-year)	Master's degree	Doctoral degree	Professional degree (JD, MD)	Total
Respondents	3	47	104	46	211	75	10	7	503
	0.60%	9.34%	20.68%	9.15%	41.95%	14.91%	1.99%	1.39%	100%

Panel D: Industry

	Agriculture, Mining, Construction, and Utilities	Finance and Real Estate	Information Technology	Manufacturing	Professional, Educational, Health, and Social Services	Transportation, Accommodation, and Food Services	Wholesale and Retail Trade	Other	Total
Respondents	29	53	126	26	97	29	80	63	503
	5.77%	10.54%	25.05%	5.17%	19.28%	5.77%	15.90%	12.52%	100%

Region E: Region

